

Product Overview

The NSI66x1A is a single-channel reinforced isolated smart gate driver to drive IGBTs and SiC MOSFETs in many applications. It can source and sink 10A peak current. System robustness is supported by 150kV/ μ s minimum common-mode transient immunity (CMTI).

The NSI66x1A includes crucial protection features such as miller clamp, DESAT, UVLO and soft turn off. UVLO and short circuit fault are reported through separate pins. ASC feature is designed to force output on in emergency, which supports system fault management.

NSI66x1A is suitable for high reliability, power density and efficiency switching power system.

Key Features

- 5.7kV_{RMS} withstand isolation voltage
- SiC MOSFETs and IGBTs up to 2121V_{pk}
- Driver side supply voltage: up to 32V with UVLO
- 10A peak source and sink output current
- High CMTI: ± 150 kV/ μ s
- 200ns fast response time of DESAT
- Monitor status of device on FLT and RDY
- 80ns typical propagation delay
- 400mA soft turn off current
- 40ns maximum pulse width distortion
- Active short circuit protection
- Operation ambient temperature: -40°C ~150°C
- RoHS & REACH Compliant
- Lead-free component, suitable for lead-free soldering profile: 260°C, MSL3
- AEC-Q100 Qualified for Grade1 : T_A from -40°C to 125°C

Safety Regulatory Approvals

- UL recognition: 5700V_{RMS}
- DIN EN IEC 60747-17(VDE 0884-17)
- CSA component notice 5A

- CQC certification per GB4943.1

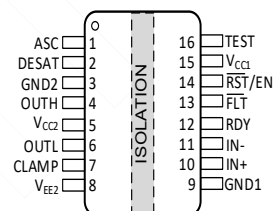
Applications

- Traction Inverter for EVs
- On-board Charger and DC/DC Converter for HEV/EVs
- UPS and Power Supplies

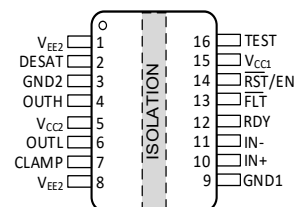
Device Information

Part Number	ASC protection	Package	Body size
NSI6611ASC-Q1SWR	Yes	SOW16	10.3×7.5×2.3mm
NSI6651ASC-Q1SWR	No	SOW16	10.3×7.5×2.3mm
NSI6651ALC-Q1SWR	No	SOW16	10.3×7.5×2.3mm

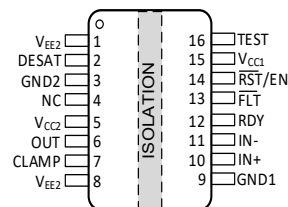
Functional Block Diagram



NSI6611ASC-Q1SWR



NSI6651ASC-Q1SWR



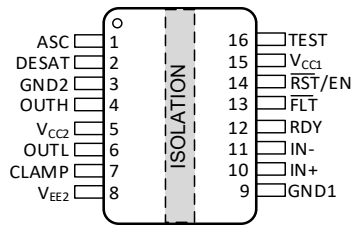
NSI6651ALC-Q1SWR

INDEX

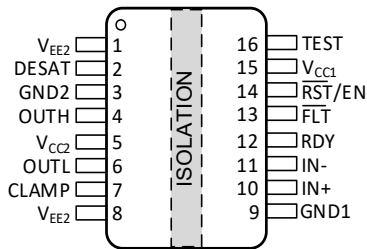
1. PIN CONFIGURATION AND FUNCTIONS	3
2. ABSOLUTE MAXIMUM RATINGS	4
3. ESD RATINGS	4
4. RECOMMENDED OPERATING CONDITIONS	4
5. THERMAL INFORMATION	5
6. SPECIFICATIONS	6
6.1. DC ELECTRICAL CHARACTERISTICS	6
DC ELECTRICAL CHARACTERISTICS(CONTINUED)	7
DC ELECTRICAL CHARACTERISTICS(CONTINUED)	9
6.2. SWITCHING ELECTRICAL CHARACTERISTICS	10
6.3. TYPICAL CHARACTERISTICS	11
7. HIGH VOLTAGE FEATURE DESCRIPTION	15
7.1. INSULATION AND SAFETY RELATED SPECIFICATIONS	15
7.2. INSULATION CHARACTERISTICS FOR SOW16 PACKAGE	16
7.3. SAFETY LIMITING VALUES FOR SOW16 PACKAGE	17
7.4. REGULATORY INFORMATION FOR SOW16 PACKAGE	18
8. FUNCTION DESCRIPTION	19
8.1. OVERVIEW	19
8.2. POWER SUPPLY	20
8.3. OUTPUT STAGE	20
8.4. V_{CC2} AND UNDER VOLTAGE LOCK OUT(UVLO)	21
8.5. ACTIVE PULL-DOWN	22
8.6. SHORT CIRCUIT CLAMPING	22
8.7. INTERNAL ACTIVE MILLER CLAMP	22
8.8. DESATURATION (DESAT) PROTECTION	24
8.9. SOFT TURN-OFF	24
8.10. FAULT(FLT AND RST/EN)	24
8.11. ASC PROTECTION(ONLY FOR NSI6611ASC)	26
8.12. DEVICE FUNCTIONAL MODES	28
9. APPLICATION NOTE	29
9.1. TYPICAL APPLICATION CIRCUIT	29
9.2. DESIGN FOR IN+, IN- AND RST/EN	29
9.3. DESIGN FOR EN/RST, RDY AND FLT	29
9.4. DESIGN FOR AUTOMATIC RESET CONTROL	30
9.5. PWM INTERLOCK PROTECTION	31
9.6. DESIGN FOR R_{ON} AND R_{OFF}	31
9.7. DESIGN FOR DESAT PROTECTION	31
9.8. DESIGN FOR EXTERNAL CURRENT BUFFER	32
9.9. PCB LAYOUT	33
10. PACKAGE INFORMATION	34
11. ORDERING INFORMATION	35
12. DOCUMENTATION SUPPORT	35
13. TAPE AND REEL INFORMATION	36
14. REVERSION HISTORY	39

1. Pin Configuration and Functions

NSI6611ASC Top View



NSI6651ASC Top View



NSI6651ALC Top View

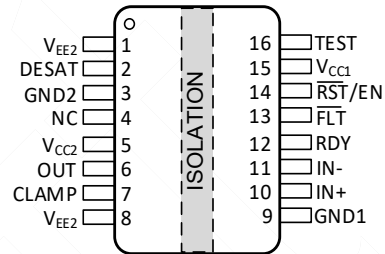


Table 1.1 NSI66x1A Pin Configuration and Description

SYMBOL	PIN NO.			FUNCTION
	NSI6611ASC	NSI6651ASC	NSI6651ALC	
ASC	1	/	/	Active Short Circuit pin. If ASC is set to high, the OUTH pin will be forced to output high under the emergency situation
DESAT	2	2	2	Fast overcurrent and short circuit protection
GND2	3	3	3	Driver side ground pin
OUTH	4	4	/	Driver source output pin
V _{CC2}	5	5	5	Driver side positive supply pin
OUTL	6	6	/	Driver sink output pin
CLAMP	7	7	7	Internal active miller clamp to prevent false turn-on
V _{EE2}	8	1,8	1,8	Driver side negative supply pin
GND1	9	9	9	Input-side ground pin
IN+	10	10	10	Non-inverting gate driver control input
IN-	11	11	11	Inverting gate driver control input
RDY	12	12	12	Power good signal. Active low to report under voltage lock
FLT	13	13	13	Fault output pin. Active low to report overcurrent or short circuit
RST	14	14	14	Enable the device if this pin is set to high ,or set to low to reset the fault signal under DESAT condition
V _{CC1}	15	15	15	Input-side power supply
NC	/	/	4	No Connection
OUT	/	/	6	Driver output pin
Test	16	16	16	Test mode pin. It is recommended to connect to GND1

2. Absolute Maximum Ratings

Parameters	Symbol	Min	Max	Unit
Driver Side Supply Voltage	$V_{CC2-V_{EE2}}$	-0.3	35	V
Driver Side Supply Voltage	$V_{CC2-GND2}$	-0.3	35	V
Driver Side Supply Voltage	$V_{EE2-GND2}$	-17.5	0.3	V
Output Signal Voltage - DC	$V_{OUTH}, V_{OUTL}, V_{CLAMP}$	$V_{EE2}-0.3$	V_{CC2}	V
Operating Junction Temperature	T_J	-40	150	°C
Storage Temperature	T_{stg}	-65	150	°C
Input Side Supply Voltage	V_{CC1}	-0.3	6	V
Input Signal Voltage - DC	$V_{IN+}, V_{IN-}, V_{RST}$	$GND1-0.3$	$V_{CC1}+0.3$	V
RDY, FLT (Input Side)	V_{RDY}, V_{FLT}	$GND1-0.3$	V_{CC1}	V
FLT and RDY input current	I_{FLT}, I_{RDY}		20	mA
DESAT (Driver Side)	V_{DESAT}	$GND2-0.3$	$V_{CC2}+0.3$	V
ASC (Driver Side)	V_{ASC}	$GND2-0.3$	$GND2+6$	V

3. ESD Ratings

		Symbol	Value	Unit
Electrostatic discharge	Human-body model (HBM), per AEC Q100-002 ⁽¹⁾	V_{ESD_HBM}	±3000	V
	Charged-device model (CDM), per AEC Q100-011	V_{ESD_CDM}	±1500	V

1) AEC Q100-002 indicates that HBM stressing shall be in accordance with the ANSI/ESDA/JEDEC JS-001 specification.

4. Recommended Operating Conditions

Parameters	Symbol	Min	Max	Unit
Ambient Temperature	T_A	-40	125	°C
Input Side Supply Voltage	$V_{CC1-GND1}$	3	5.5	V
Driver Side Supply Voltage	$V_{CC2-GND2}$	13	32	V
Driver Side Supply Voltage	$V_{CC2-V_{EE2}}$	-	32	V
ASC (Driver Side)	V_{ASC}	$GND2$	$GND2+5$	V
$IN+$, $IN-$, $\overline{RST}/EN$ (Respect to $GND1$)	High level input voltage	$0.7 \times V_{CC1}$	V_{CC1}	V
	Low level input voltage	0	$0.3 \times V_{CC1}$	

5. Thermal Information

Parameters	Symbol	SOW16	Unit
Junction-to-ambient thermal resistance	$R_{\theta JA}$	97.0	°C/W
Junction-to-top characterization parameter	Ψ_{JT}	35.8	°C/W
Junction-to-board characterization parameter	Ψ_{JB}	39.0	°C/W
Junction-to-case(top) thermal resistance	$R_{JC(top)}$	23.3	°C/W

- 1) Standard JESD51-3 Low Effective Thermal Conductivity Test Board (1s) in an environment described in JESD51-2a.
- 2) Standard JESD51-3 Low Effective Thermal Conductivity Test Board (1s) by transient dual interface test method described in JESD51-14.
- 3) Obtained by Simulating in an environment described in JESD51-2a.

6. Specifications

6.1. DC Electrical Characteristics

All min and max specifications are at $T_A = -40^{\circ}\text{C}$ to 150°C . Typical values are tested at $V_{CC1} = 3.3\text{V}$ or 5V , $V_{CC2} = 15\text{V}$ or 30V , $V_{EE2} = \text{GND2}$.

Parameter	Symbol	Min	Typ	Max	Unit	Condition
Input Side Supply						
Input side Supply Quiescent Current	I _{CC1}	0.6	1.6	4	mA	V _{CC1} =5V,OUT=High
V _{CC1} UVLO Rising Threshold	V _{CC1_ON}	2.5	2.7	2.9	V	
V _{CC1} UVLO Falling Threshold	V _{CC1_OFF}	2.3	2.5	2.7	V	
V _{CC1} UVLO Hysteresis	V _{CC1_HYS}		0.2		V	
V _{CC1} UVLO deglitch time	t _{VCC1_FIL⁽¹⁾}		10		μs	
V _{CC1} UVLO on delay to output high	t _{VCC1H_OU_T⁽¹⁾}		33			IN+=V _{CC1} IN- =GND1
V _{CC1} UVLO off delay to output low	t _{VCC1L_OUT⁽¹⁾}		10			
V _{CC1} UVLO on delay to RDY high	t _{VCC1H_RD_Y⁽¹⁾}		38			RST/EN=V _{CC1}
V _{CC1} UVLO off delay to RDY low	t _{VCC1L_RDY⁽¹⁾}		10			
Driver Side Supply						
Driver side Supply Quiescent Current	I _{CC2}	1	3.3	7	mA	V _{CC2} =15V, V _{EE2} =GND2, OUT=High
V _{CC2} UVLO Rising Threshold	V _{CC2_ON}	9.8	11.2	12.8	V	
V _{CC2} UVLO Falling Threshold	V _{CC2_OFF}	9.0	10.4	11.8	V	
V _{CC2} UVLO Hysteresis	V _{CC2_HYS}		0.8		V	
V _{CC2} UVLO deglitch time	t _{VCC2_FIL⁽¹⁾}		5		μs	
V _{CC2} UVLO on delay to output high	t _{VCC2H_OU_T⁽¹⁾}		11			IN+=V _{CC1} IN- =GND1
V _{CC2} UVLO off delay to output low	t _{VCC2L_OUT⁽¹⁾}		10			
V _{CC2} UVLO on delay to RDY high	t _{VCC2H_RD_Y⁽¹⁾}		11			RST/EN=V _{CC1} , DESAT=GND2
V _{CC2} UVLO off delay to RDY low	t _{VCC2L_RDY⁽¹⁾}		14			
RDY Reporting						
V _{CC2} UVLO RDY low minimum holding time	t _{RDY_HLD}			1	ms	

Parameter	Symbol	Min	Typ	Max	Unit	Condition
Open drain low output voltage	V_{RDY_L}			0.3	V	$I_{SINK_RDY}=5mA$
Input Pin Characteristic						
Logic High Input Threshold (IN+, IN-, RST)	V_{INH}	2.5	2.9	3.5	V	$V_{CC1}=5V$
Logic Low Input Threshold (IN+, IN-, RST)	V_{INL}	1.5	2.1	2.5	V	$V_{CC1}=5V$
Input Hysteresis Voltage (IN+, IN-, RST)	V_{hys_IN}		0.8		V	$V_{CC1}=5V$
IN+ Input Current	I_{IN+_H}		90		μA	$V_{IN+}=5V$

DC Electrical Characteristics(continued)

All min and max specifications are at $T_A=-40^{\circ}C$ to $150^{\circ}C$. Typical values are tested at $V_{CC1}=3.3V$ or $5V$, $V_{CC2}=15V$ or $30V$, $V_{EE2}=GND2$.

Parameter	Symbol	Min	Typ	Max	Unit	Condition
IN- Input Current	I_{IN-_L}		-90		μA	$V_{IN-}=GND1$
ASC Pin Characteristic (only for NSI6611ASC)						
Logic High Input Threshold (ASC)	V_{ASCH}	2.7	2.9	3.2	V	
Logic Low Input Threshold (ASC)	V_{ASCL}	1.3	1.5	1.7	V	
ASC to output rising edge delay	t_{ASC_r}	390	560	1100	ns	
ASC to output falling edge delay	t_{ASC_f}	150	360	480	ns	
Enable Pin Characteristic						
RST deglitch filter time for Enable/Shutdown	t_{min_RST}	28	40	60	ns	
Output Pin Characteristic						
High Level Output Voltage	V_{OH}		$V_{CC2}-0.6$		V	$I_{OUT}=-200mA$, $V_{IN+}=High$, $V_{IN-}=Low$
Low Level Output Voltage	V_{OL}		0.1		V	$I_{OUT}=200mA$, $V_{IN+}=Low$, $V_{IN-}=Low$
Output pull-up resistance	R_{OH}		2.2		Ω	$I_{OUT}=-0.1A$, $V_{IN+}=High$, $V_{IN-}=Low$
Output pull-down resistance	R_{OL}		0.3		Ω	$I_{OUT}=0.1A$, $V_{IN+}=Low$, $V_{IN-}=High$
Low Level Clamp Voltage	V_{CLAMP}		$V_{EE2}+0.8$		V	$I_{CLAMP}=1A$, $V_{IN+}=Low$, $V_{IN-}=Low$
High Level Peak Output Current	$I_{OUTH}^{(1)}$		11		A	$V_{CC2}=15V$, pulse width<10us
Low Level Peak Output Current	$I_{OUTL}^{(1)}$		12		A	
			8		A	$V_{OUT}=V_{EE2}+2.5V$

Clamp Threshold Voltage	$V_{CLAMP-TH}$	1.5	2	2.5	V	V_{CLAMP} falling, $V_{IN+}=Low$, $V_{IN-}=Low$
Calmp Delay	$t_{DCLMP}^{(2)}$		70		ns	
OUT Short Circuit Clamping Voltage	V_{CLP_OUT}		$V_{CC2}+1.1$	$V_{CC2}+2$	V	$V_{IN+}=High$, $V_{IN-}=Low$, $I_{OUTL}=0.5A$, pulse width<10us
CLAMP Short Circuit Clamping Voltage	V_{CLP_CLAMP}		$V_{CC2}+1.8$	$V_{CC2}+2.5$	V	$V_{IN+}=High$, $V_{IN-}=Low$, $I_{OUTH}=0.5A$, pulse width<10us
			$V_{CC2}+0.8$		V	$V_{IN+}=High$, $V_{IN-}=Low$, $I_{CLAMP}=20mA$
OUT Active Pull-Down Voltage	V_{SD_OUT}	1.5	2.3	3.1	V	$V_{CC2}=OPEN$, $I_{OUTL}=0.1 \times I_{OUTL}(typ)$

DC Electrical Characteristics(continued)

All min and max specifications are at $T_A = -40^{\circ}\text{C}$ to 150°C . Typical values are tested at $V_{CC1} = 3.3\text{V}$ or 5V , $V_{CC2} = 15\text{V}$ or 30V , $V_{EE2} = \text{GND2}$.

Parameter	Symbol	Min	Typ	Max	Unit	Condition
Desaturation						
Blanking Capacitor Discharge Current	I_{DCHG}	10	15		mA	$V_{\text{DESAT}} = 6\text{V}$
Blanking Capacitor Charge Current	I_{CHG}	350	500	650	μA	$V_{\text{DESAT}} = 2\text{V}$
Detection Threshold	$V_{\text{DESAT_TH}}$	8.5	9.3	10	V	
Leading edge blank time	$t_{\text{DESAT_LEB}}^{(1)}$		200		ns	
DESAT deglitch filter time	$t_{\text{DESAT_FIL}}$	100	200	320	ns	
DESAT sense to OUT(L) 90% delay	$t_{\text{DESAT_OFF}}$	150	250	360	ns	
DESAT sense to FLT low delay	$t_{\text{DESAT_FLT}}$	400	650	800	ns	
FLT mute time	$t_{\text{FLT_MUTE}}$	0.55		1.3	ms	
RST deglitch filter time for Reseting FLT	$t_{\text{RST_FIL}}$	480	600	800	ns	
Soft turn off						
Soft turn off current	I_{STO}	100	400	570	mA	
FLT Reporting						
Open drain low output voltage	$V_{\text{FLT_L}}$			0.3	V	$I_{\text{SINK_FLT}} = 5\text{mA}$

⁽¹⁾ Not test covered, guaranteed by characterization.

⁽²⁾ Not test covered, guaranteed by design.

6.2. Switching Electrical Characteristics

Typical values are at $V_{CC1}=5V$, $V_{CC2}=15V$, $V_{EE2}=GND2$. All min and max specifications are at $T_A=-40^{\circ}C$ to $150^{\circ}C$, $C_L=0.1nF$

Parameter	Symbol	Min	Typ	Max	Unit	Condition
Minimum Pulse Width	$t_{PWmin_IN+,\ IN-}$	20	40	100	ns	
Output Rise Time	t_R		56		ns	$C_{LOAD}=10nF$
Output Fall Time	t_F		53		ns	
Propagation Delay	$t_{pLH_IN+,\ IN-}$	40	85	130	ns	$C_{LOAD}=0.1nF$
	$t_{pHL_IN+,\ IN-}$	40	80	130	ns	
Pulse Width Distortion $ t_{pHL}-t_{pLH} $	t_{PWD}			50	ns	
Common Mode Transient Immunity	$CMTI^{(1)}$	150			$kV/\mu s$	

⁽¹⁾ Not test covered, guaranteed by characterization.

⁽²⁾ Not test covered, guaranteed by design.

6.3. Typical Characteristics

$V_{CC1}=5V$, $V_{CC2}=15V$, $V_{EE2}=GND2$ for NSI66x1A. Output has no load unless otherwise noted.

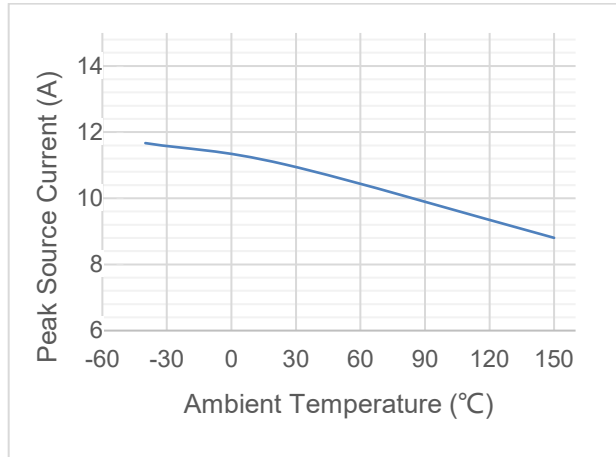


Figure 6.1 I_{peak_source} vs Temperature

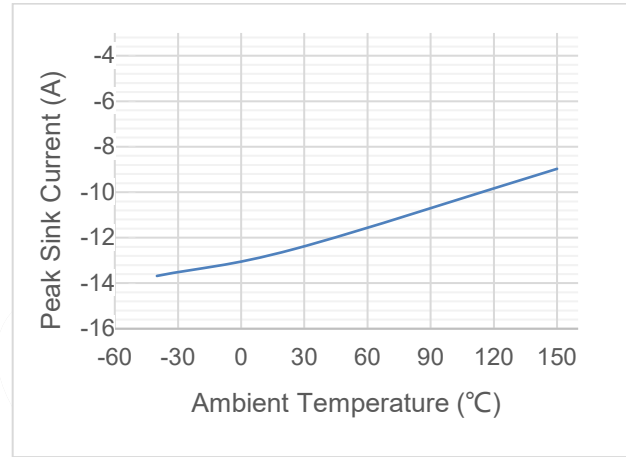


Figure 6.2 I_{peak_sink} vs Temperature

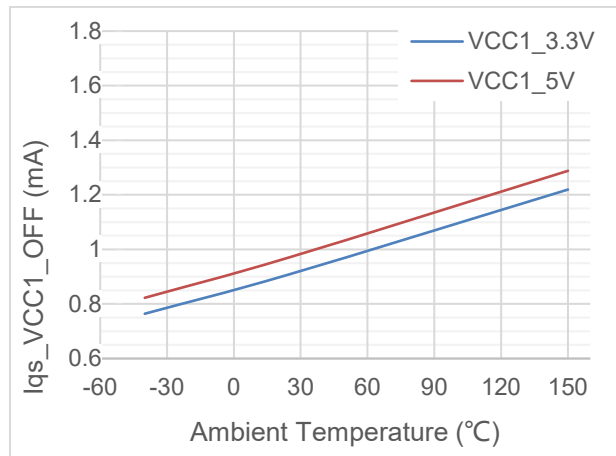


Figure 6.3 $I_{qs_VCC1_OFF}$ vs Temperature

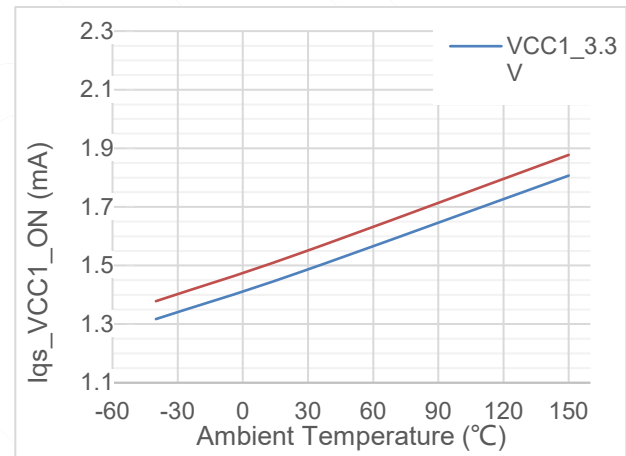


Figure 6.4 $I_{qs_VCC1_ON}$ vs Temperature

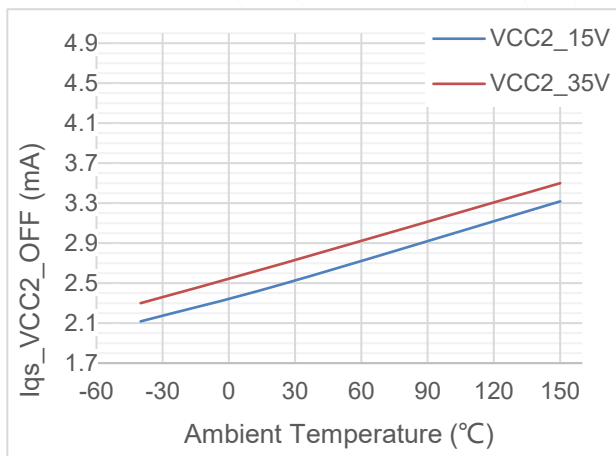


Figure 6.5 $I_{qs_VCC2_OFF}$ vs Temperature

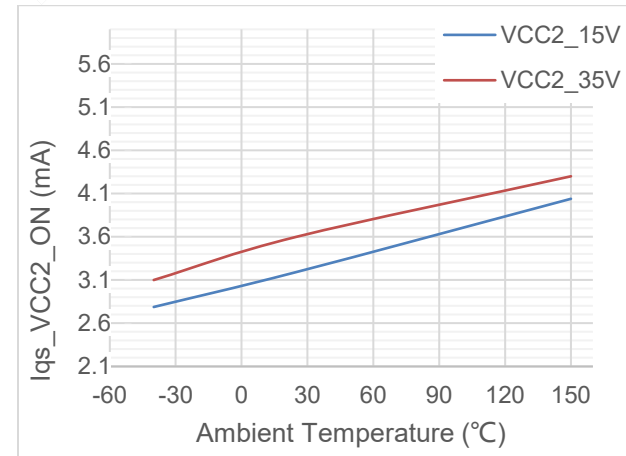


Figure 6.6 $I_{qs_VCC2_ON}$ vs Temperature

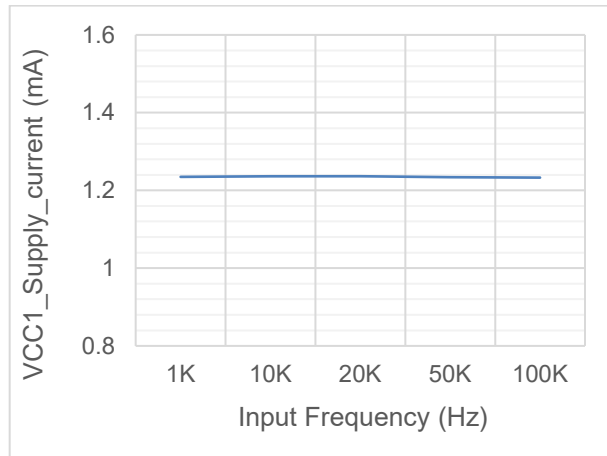


Figure 6.7 V_{CC1} Supply current vs Input Frequency

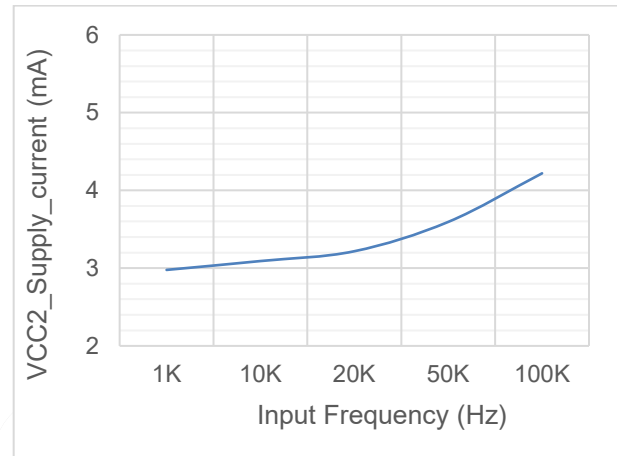


Figure 6.8 V_{CC2} Supply current vs Input Frequency

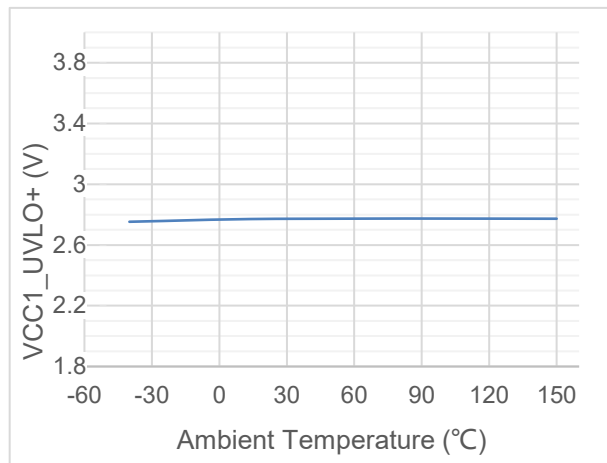


Figure 6.9 V_{CC1} UVLO+ vs Temperature

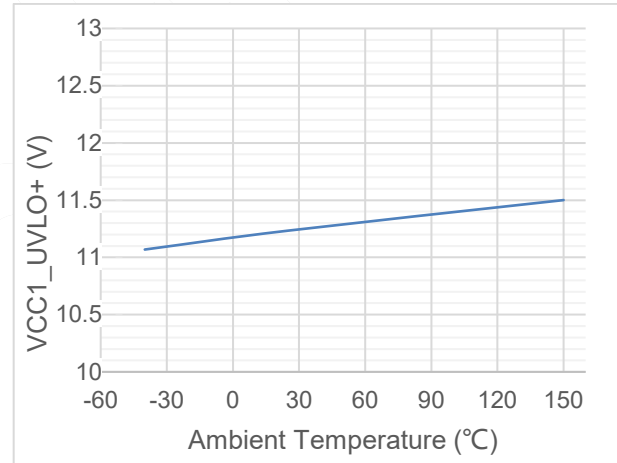


Figure 6.10 V_{CC2} UVLO+ vs Temperature

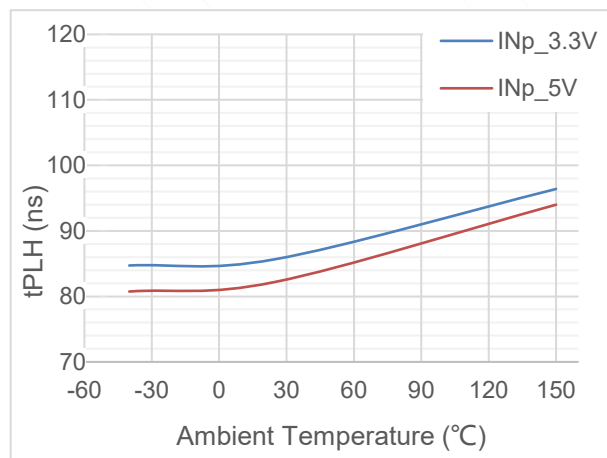


Figure 6.11 t_{PLH} vs Temperature

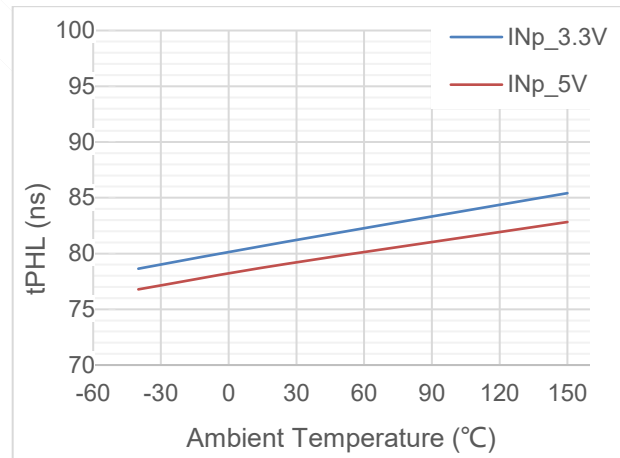


Figure 6.12 t_{PHL} vs Temperature

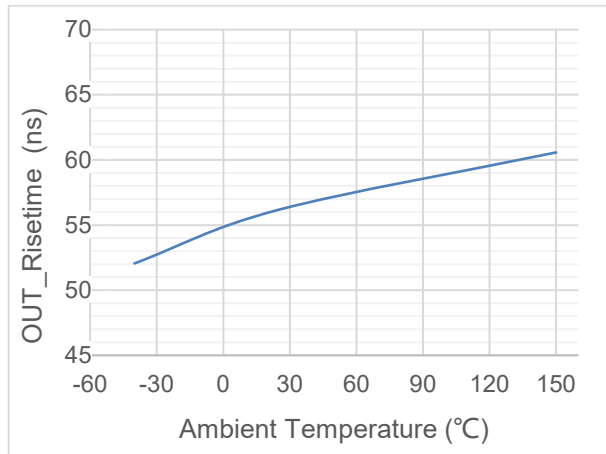


Figure 6.13 OUT_Risetime vs Temperature

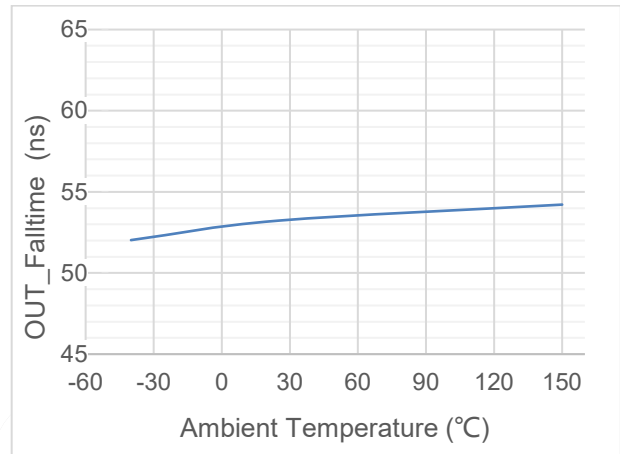


Figure 6.14 OUT_Falltime vs Temperature

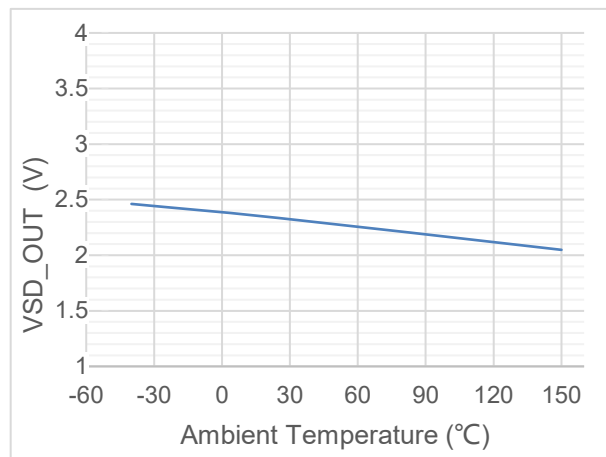


Figure 6.15 VSD_OUT vs Temperature

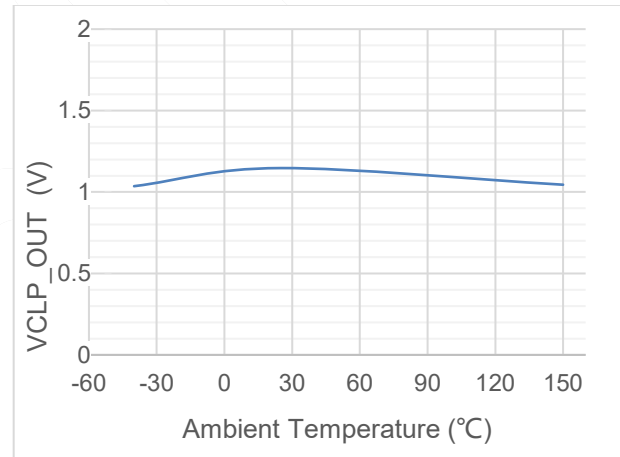


Figure 6.16 VCLP_OUT vs Temperature

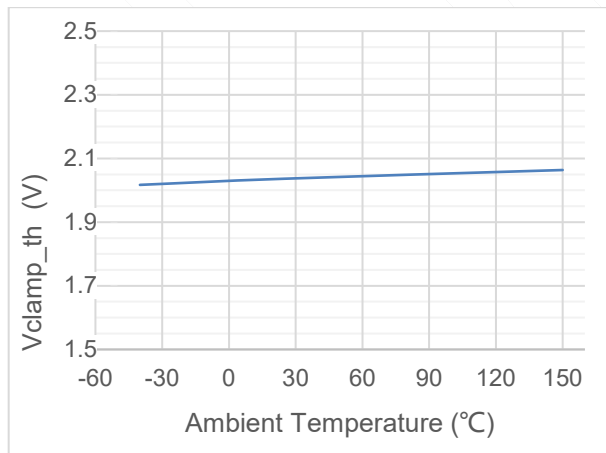


Figure 6.17 Vclamp_th vs Temperature

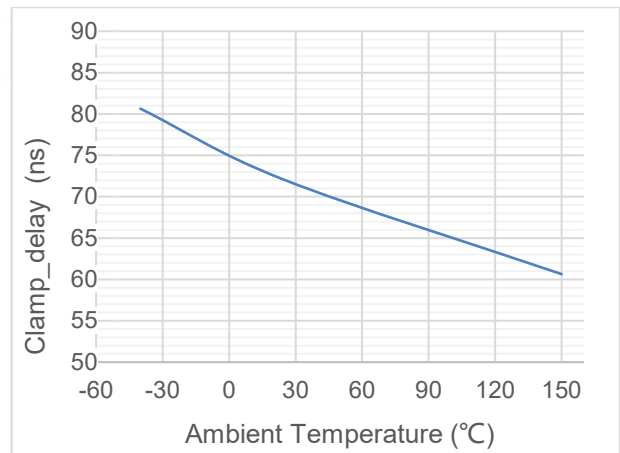


Figure 6.18 Clamp_delay vs Temperature

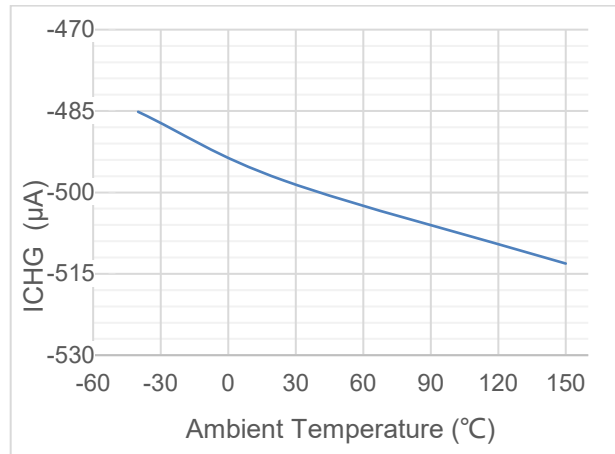


Figure 6.19 I_CHG vs Temperature

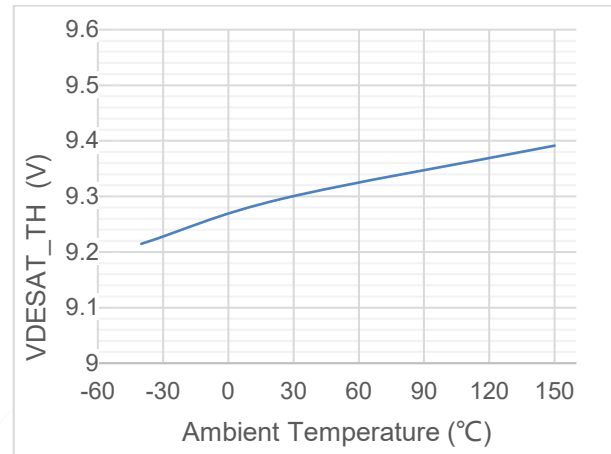


Figure 6.20 DESAT Threshold Voltage vs Temperature

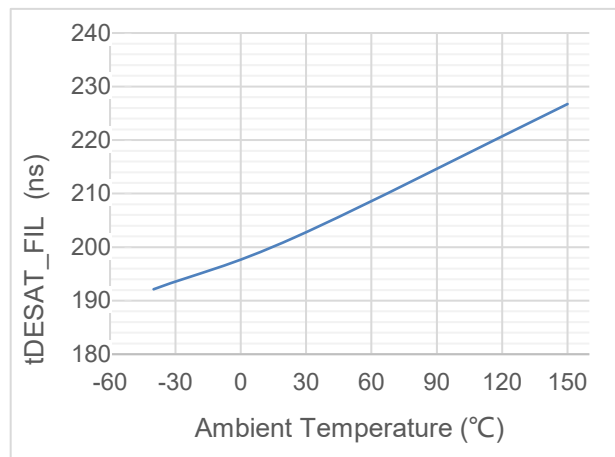


Figure 6.21 Deglitch time vs Temperature

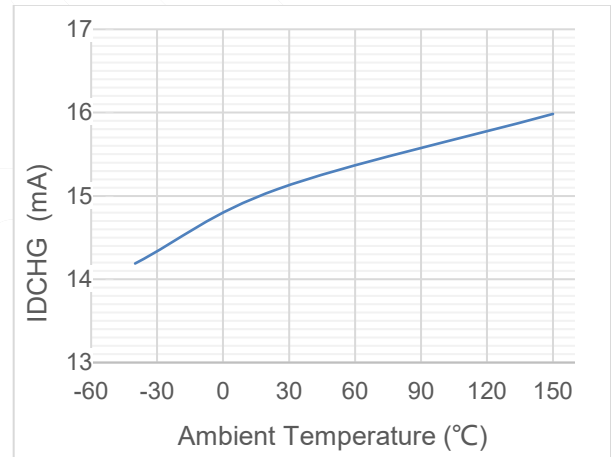


Figure 6.22 I_DCHG vs Temperature

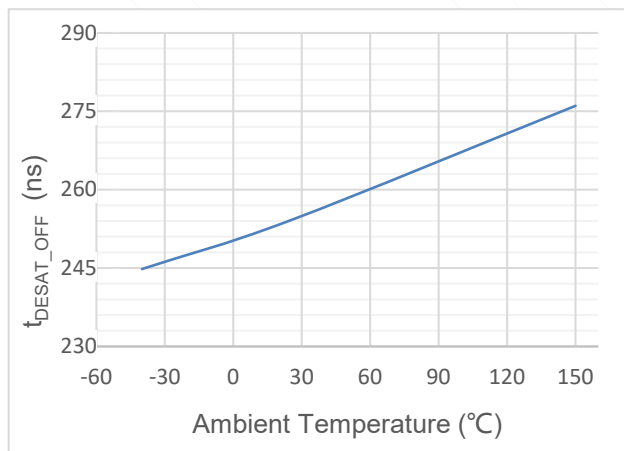


Figure 6.23 t_DESAT_OFF vs Temperature

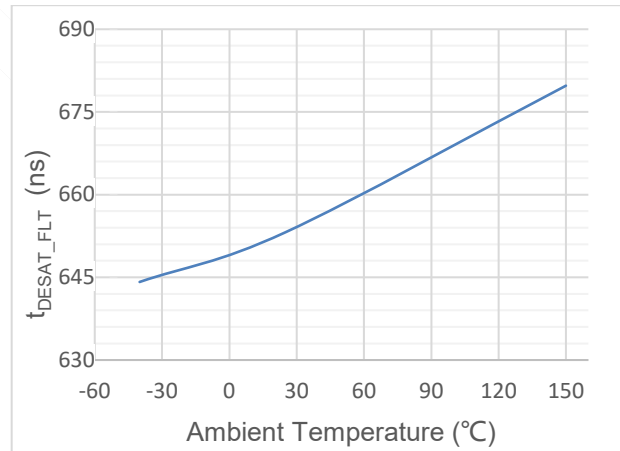


Figure 6.24 t_DESAT_FLT vs Temperature

7. High Voltage Feature Description

7.1. Insulation and Safety Related Specifications

Parameter	Symbol	SOW16	Unit	Comments
Minimum External Clearance	CLR	8.0	mm	IEC 60664-1
Minimum External Creepage	CPG	8.0	mm	IEC 60664-1
Distance Through Insulation	DTI	20	µm	Minimum internal gap
Tracking Resistance (Comparative Tracking Index)	CTI	>600	V	DIN EN 60112 (VDE 0303-11); IEC 60112
Material Group		I		

Description	Test Condition	Value
Overvoltage Category per IEC60664-1	For Rated Mains Voltage $\leq 150\text{Vrms}$	I to IV
	For Rated Mains Voltage $\leq 300\text{Vrms}$	I to IV
	For Rated Mains Voltage $\leq 600\text{Vrms}$	I to IV
	For Rated Mains Voltage $\leq 1000\text{Vrms}$	I to III
Climatic Classification		40/125/21
Pollution Degree per DIN VDE 0110		2

7.2. Insulation Characteristics for SOW16 Package

Description	Test Condition	Symbol	Value	Unit
Maximum Working Isolation Voltage	AC voltage (sine wave) Time dependent dielectric breakdown (TDDB) test	V_{IOWM}	1500	V_{RMS}
	DC voltage		2121	V_{DC}
Maximum Repetitive Peak Isolation Voltage	AC voltage (bipolar)	V_{IORM}	2121	V_{PEAK}
Apparent Charge	Method a, after Input/output safety test subgroup 2/3, $V_{ini}=V_{IOTM}$, $t_{ini} = 60s$, $V_{pd}(m)=1.2 \times V_{IORM}$, $t_m=10s$.	q_{pd}	<5	pC
	Method a, after environmental tests subgroup 1, $V_{ini}=V_{IOTM}$, $t_{ini}=60s$, $V_{pd}(m)=1.6 \times V_{IORM}$, $t_m=10s$			
	Method b, routine test (100% production) and preconditioning (type test); $V_{ini}=1.2 \times V_{IOTM}$, $t_{ini}=1s$ $V_{pd}(m)=1.875 \times V_{IORM}$, $t_m=1s$ (method b1) or $V_{pd}(m)=V_{ini}$, $t_m=t_{ini}$ (method b2)			
Maximum Transient Isolation Voltage	$t = 60s$	V_{IOTM}	8000	V_{PEAK}
Maximum impulse voltage	Tested in air, 1.2/50-us waveform per IEC62368-1	V_{IMP}	6250	V_{PEAK}
Maximum Surge Isolation Voltage	Test method per IEC60065, 1.2/50us waveform, $V_{TEST}=V_{IOSM} \times 1.6$	V_{IOSM}	10000	V_{PEAK}
Isolation Resistance	$V_{IO} = 500V$ at $T_A=T_S$	R_{IO}	$>10^9$	Ω
	$V_{IO} = 500V$ at $100^\circ C \leq T_A \leq 125^\circ C$		$>10^{11}$	Ω
	$V_{IO} = 500V$, $T_A=25^\circ C$		$>10^{12}$	Ω
Isolation Capacitance	$f = 1MHz$	C_{IO}	0.8	pF
UL1577				
Maximum Withstanding Isolation Voltage	$V_{TEST}= V_{ISO}$, $t = 60s$ (qualification); $V_{TEST}= 1.2 \times V_{ISO}$, $t = 1s$ (100% production)	V_{ISO}	5700	V_{RMS}

7.3. Safety Limiting Values for SOW16 Package

Description	Test Condition	Symbol	Value		Unit
Maximum Safety Temperature		T_s	150		°C
Maximum Safety Power Dissipation	$R_{\theta JA}=97^{\circ}\text{C/W}$, $T_J=150^{\circ}\text{C}$, $V_{CC2}=20\text{V}$, $V_{EE2}=-5\text{V}$ $T_A=25^{\circ}\text{C}$	P_s	Total	1288	mW
Maximum Safety Current	$R_{\theta JA}=97^{\circ}\text{C/W}$, $V_{CC2}=15\text{V}$, $V_{EE2}=-5\text{V}$ $T_J=150^{\circ}\text{C}$, $T_A=25^{\circ}\text{C}$	I_s	Driver side	64.4	mA
	$R_{\theta JA}=97^{\circ}\text{C/W}$, $V_{CC2}=20\text{V}$, $V_{EE2}=-5\text{V}$ $T_J=150^{\circ}\text{C}$, $T_A=25^{\circ}\text{C}$		Driver side	51.5	

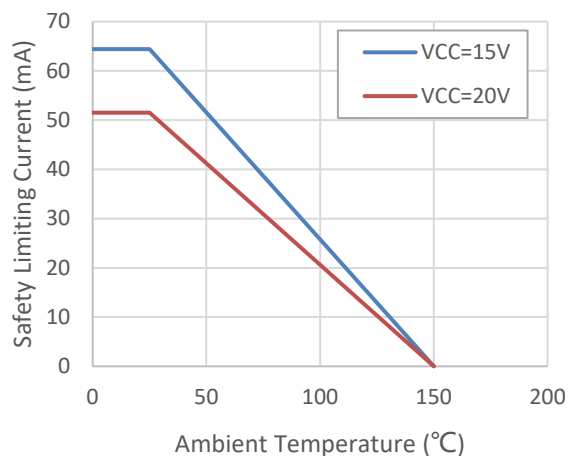


Figure 7.1 Thermal Derating Curve for Limiting Current per DIN VDE V 0884-11 for SOW16 Package

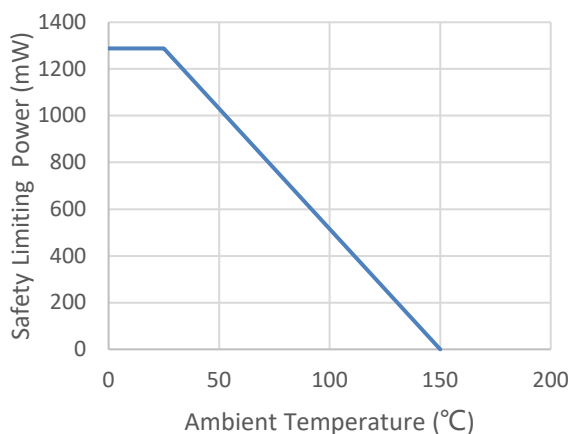


Figure 7.2 Thermal Derating Curve for Limiting Power per DIN VDE V 0884-11 for SOW16 Package

7.4. Regulatory Information for SOW16 Package

<i>UL</i>		<i>VDE</i>	<i>CQC</i>
UL 1577 Component Recognition Program	Approved under CSA Component Acceptance Notice 5A	DIN EN IEC 60747-17 (VDE 0884-17)	Certified according to GB4943.1
Single Protection, 5700V _{RMS} Isolation Voltage	Single Protection, 5700V _{RMS} Isolation voltage	Reinforced Insulation $V_{IORM}=2121V_{PEAK}$, $V_{IOTM}=8000V_{PEAK}$, $V_{IOSM}=10000V_{PEAK}$	Reinforced Insulation
E500602		40052820	CQC20001264939

8. Function Description

8.1. Overview

The NSI66x1A is a high reliable power transistor gate driver. It can source and sink 10A peak current, which is suitable to drive MOSFET, IGBT, or SiC MOSFET. The NSI66x1A is available in SOW16 package, which can support 5700V_{RMS} isolation per UL1577. System robustness is supported by 150kV/us minimum common-mode transient immunity (CMTI).

Besides, the NSI66x1A includes crucial protection features such as miller clamp, DESAT short circuit detection and soft turn off. UVLO and short circuit fault are reported through separate pins. ASC feature is designed to force output ON in emergency situation, which supports system fault management.

The isolation barrier inside NSI66x1A is based on a capacitive isolation. The modulation uses OOK modulation technique with key benefits of high noise immunity and low radiation EMI. The digital signal is modulated with RF carrier generated by the internal oscillator at the transmitter side, then it is transferred through the capacitive isolation barrier and demodulated at the receiver side.

The functional block diagram of NSI66x1A is shown in Figure 8.1. Two Input pins with non-inverting and inverting logic support interlock and shoot through protection. Low resistance of high side and low side MOSFET in the output stage ensures high driving capability. Split outputs can control the rise and fall time individually. Active pull-down and short circuit clamping features are implemented to protect power transistor.

In summary, the NSI66x1A is suitable to replace source and sink reinforced gate driver in high reliability, power density and efficiency switching power system.

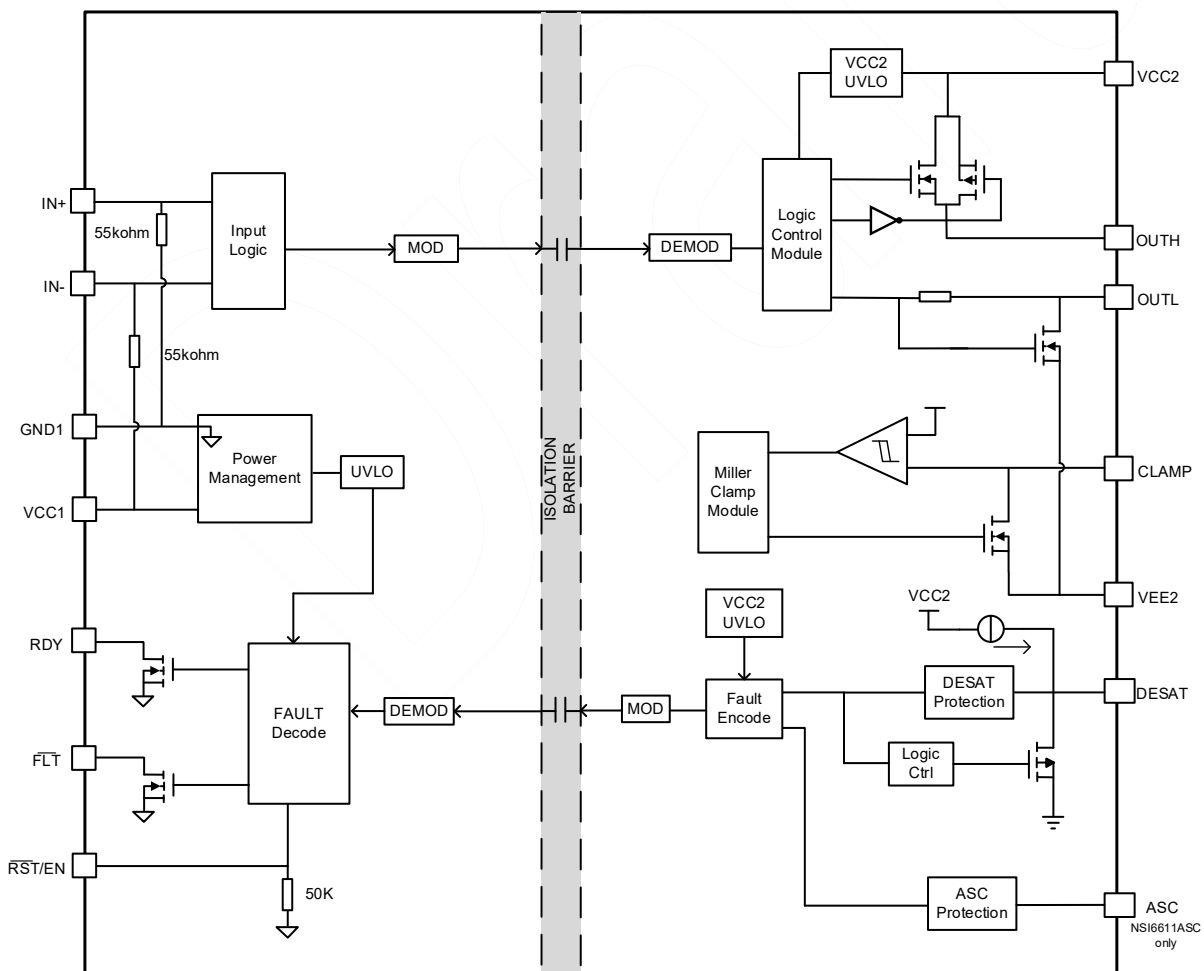


Figure 8.1 NSI66x1A Function Block Diagram

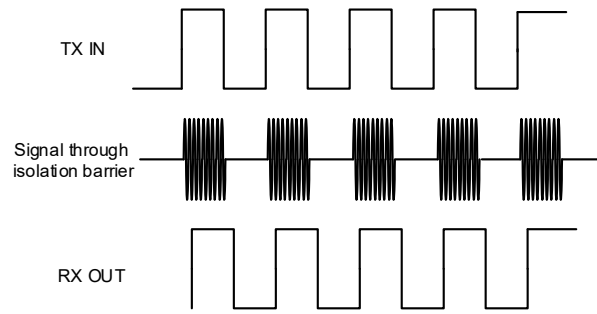


Figure 8.2 OOK Modulation

8.2. Power Supply

Power supply V_{CC1} is able to support from 3.3V to 5.5V. Power supply V_{CC2} is able to support from 13V to 32V. To be mentioned, NSI66x1A also supports bipolar power supply mode on the driver side. In the case of fast switching speed, the negative power supply is crucial to prevent false turn on from parasitic Miller capacitor.

8.3. Output Stage

The NSI66x1A has P-channel and N-channel MOSFET in parallel to pull up the OUTH pin when turning on external power transistor. During DC measurement, only the P-channel MOSFET is conducting. The measurement result R_{OH} represents the on-resistance of P-channel MOSFET. The voltage and current of external power transistor drain to source or collector to emitter change during turn on. At that time, the NSI66x1A N-channel MOSFET turns on to pull up OUTH more quickly. It results external power transistor faster turn on time, lower turn on power loss, also leads to lower temperature increase of NSI66x1A. The equivalent pull-up resistance of NSI66x1A is the parallel combination $R_{OH} \parallel R_{NMOS}$. The result is quite small, indicating the strong driving capability of NSI66x1A. The pull-down structure of NSI66x1A is simply composed of an N-channel MOSFET with on-resistance of R_{OL} . The result is quite small, indicating the strong driving capability of NSI66x1A.

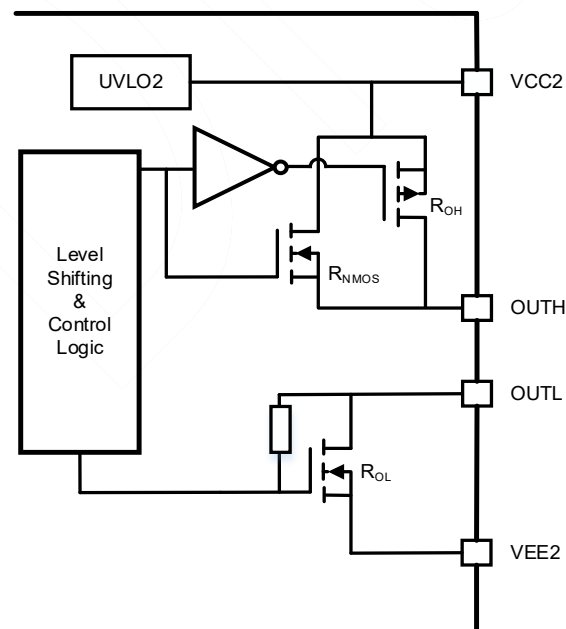


Figure 8.3 NSI66x1A output stage

8.4. V_{CC2} and Under Voltage Lock Out(UVLO)

To ensure correct switching NSI66x1A is equipped with an under voltage lockout for input and output power supply independently. V_{CC1} voltage should not fall below the UVLO threshold for normal operation, or the gate-driver output can become clamped low. Output supply UVLO is referred to GND2 pin. If V_{CC2} -GND2 falls below the UVLO threshold, OUTL of the gate-driver will be clamped low.

Local bypass capacitors should be placed between the V_{CC2} and GND2 pins, as well as the V_{CC1} and GND1 pins. 220nF to 10 μ F is recommended for device biasing. Additional 100nF capacitor in parallel with the device biasing capacitor is recommended for high frequency filtering. The capacitors should be positioned as close to the device as possible for better noise filtering. Low-ESR, ceramic surface-mount capacitors are recommended. The RDY pin will report a power good signal if the device is out of UVLO condition.

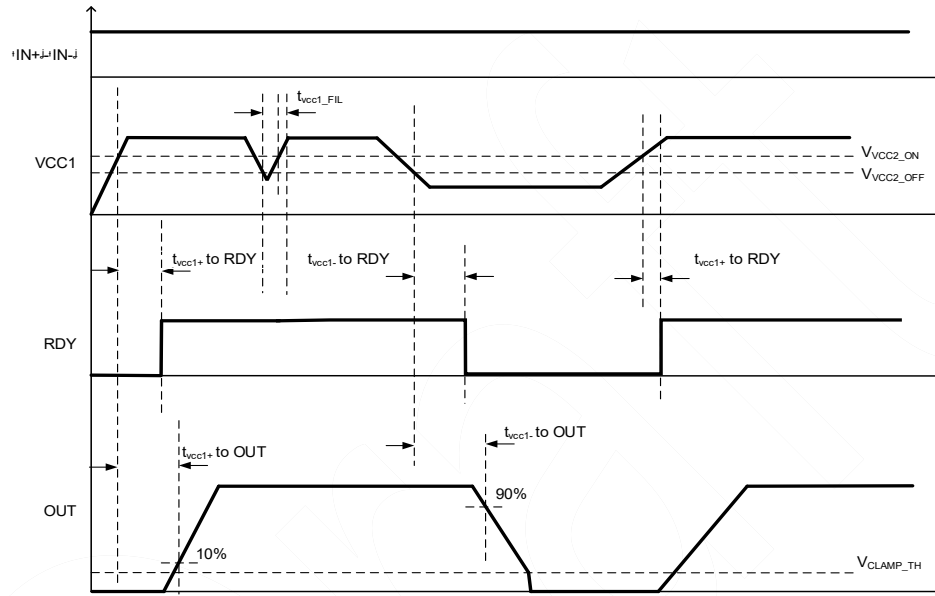


Figure 8.4 RDY vs V_{CC1} timing Diagram

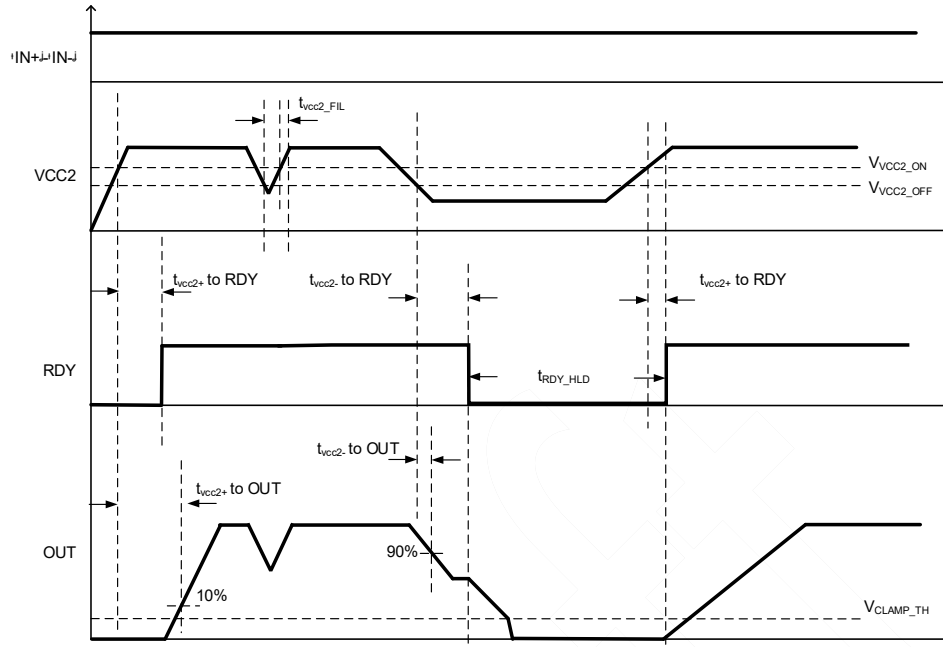


Figure 8.5 RDY vs V_{CC2} timing Diagram

8.5. Active Pull-Down

The Active Pull-Down feature ensures a safe IGBT or MOSFET off-state if V_{CC2} is not connected to the power supply. When V_{CC2} is floating, the driver output is held low and clamping OUT to approximately 1.8V higher than V_{EE2} .

8.6. Short Circuit Clamping

During short circuit the gate voltage of IGBT or MOSFET tends to rise because of the feedback via the miller capacitance. The diode between OUTH/CLAMP and V_{CC2} pins inside the driver limits this voltage to approximately 0.7V higher than the supply voltage. A maximum current of 500mA may be fed back to the supply through this path for 10 μ s. If higher currents are expected or tighter clamping is desired external Schottky diodes may be added.

8.7. Internal Active Miller Clamp

Active miller clamp is used to prevent false turn on. After the external power transistor is turned off, the other one of the phase leg is turned on. The voltage of the drain-source or collector-emitter rises instantly. The dv/dt will cause a high current on miller parasitic capacitor. The voltage drop on the gate resistor possibly turn on the external power transistor unintentionally, which will cause a catastrophic damage. To deal with that, NSI66x1A is equipped with a miller clamp pin. The clamp pin detects the gate voltage of IGBT or MOSFET. When the gate voltage is decreasing and reaches the V_{CLAMP_TH} , the clamp pin will be pulled down by the internal MOSFET, providing a low impedance path to avoid the false turn on. To be mentioned, the V_{CLAMP_TH} is 2V higher than V_{EE2} . In the situation of fast switching speed, The negative power supply is necessary to avoid false turn on.

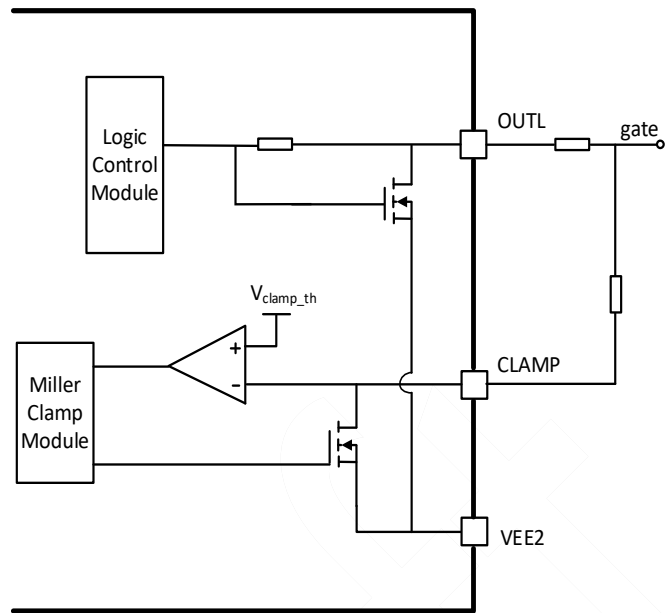


Figure 8.6 Active Miller Clamp

8.8. Desaturation (DESAT) Protection

Desaturation protection is used to prevent the power transistor from short circuit. The DESAT pin has a typical 9V threshold, which means the output will be driven low if DESAT pin reaches 9V. By default, the DESAT pin is pulled down by internal MOSFET. The internal 500 μ A current source is designed to work only when the output is high level. There is a 200ns leading edge blanking time to filter the overshoot when the external power transistor is turned on. The current source begin to charge after the internal leading edge blanking time.

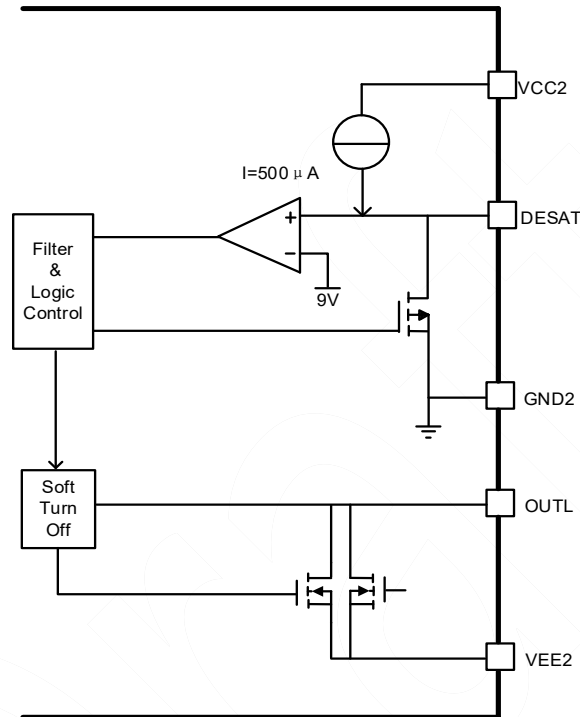


Figure 8.7 DESAT Protection

8.9. Soft Turn-off

The soft turn-off is designed to prevent the overshoot breakdown when DESAT protection is triggered. When the short circuit fault occurs, the external power transistor transits from the active zone to ohmic zone very quickly. The high di/dt may result in the overshoot voltage on the parasitic inductance of the emitter. Therefore, the device should be turned off in a soft manner. But the turn off speed should not be too slow. There is a balance between the overshoot and large energy dissipation. 400mA soft turn off current is a compromising choice.

8.10. Fault (FLT and RST/EN)

The FLT pin of NSI66x1A is used to report a warning signal if the fault is detected on DESAT. If the fault occurs, the FLT pin will be pulled down and held in low state for a mute time. During the mute time, NSI66x1A ignores any reset signal. After that, the FLT pin will be reset to high impedance status if the reset signal is checked. The RST/EN pin is pulled down to GND1 by an internal resistor, which means the device is disabled by default. Therefore, the RST/EN pin must be pulled up externally to enable the device. Timing diagram of fault report is shown below.

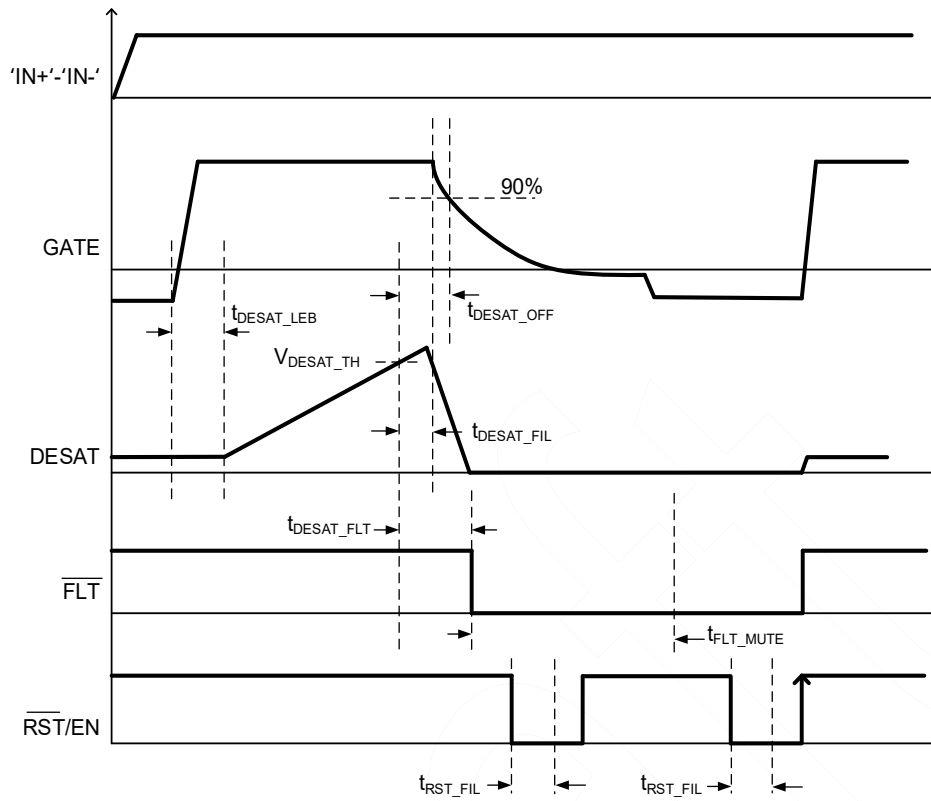


Figure 8.8 DESAT protection Timing Diagram

8.11.ASC Protection(only for NSI6611ASC)

If the active short circuit (ASC) pin is set to be high, the output will be high no matter how input-side configuration. To be mentioned, the priority of ASC protection is higher than the V_{CC1} UVLO but lower than the V_{CC2} UVLO and DESAT protection. To be mentioned, If V_{CC1} is open, DESAT protection will not be allowed. So the ASC is usable without DESAT protection.

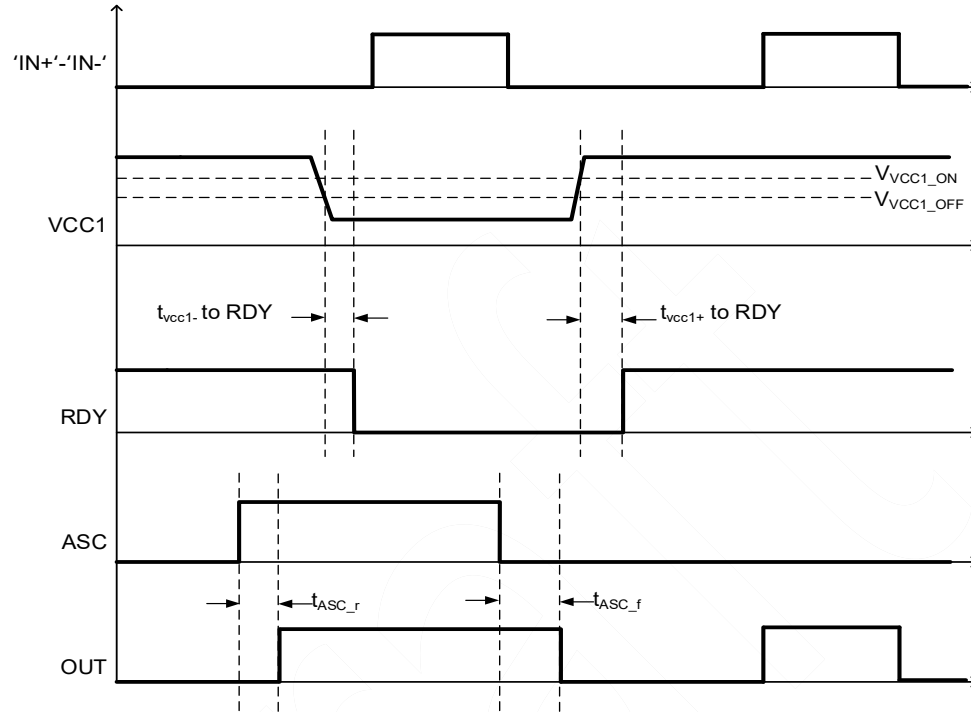


Figure 8.9 ASC protection with V_{CC1} UVLO Timing Diagram

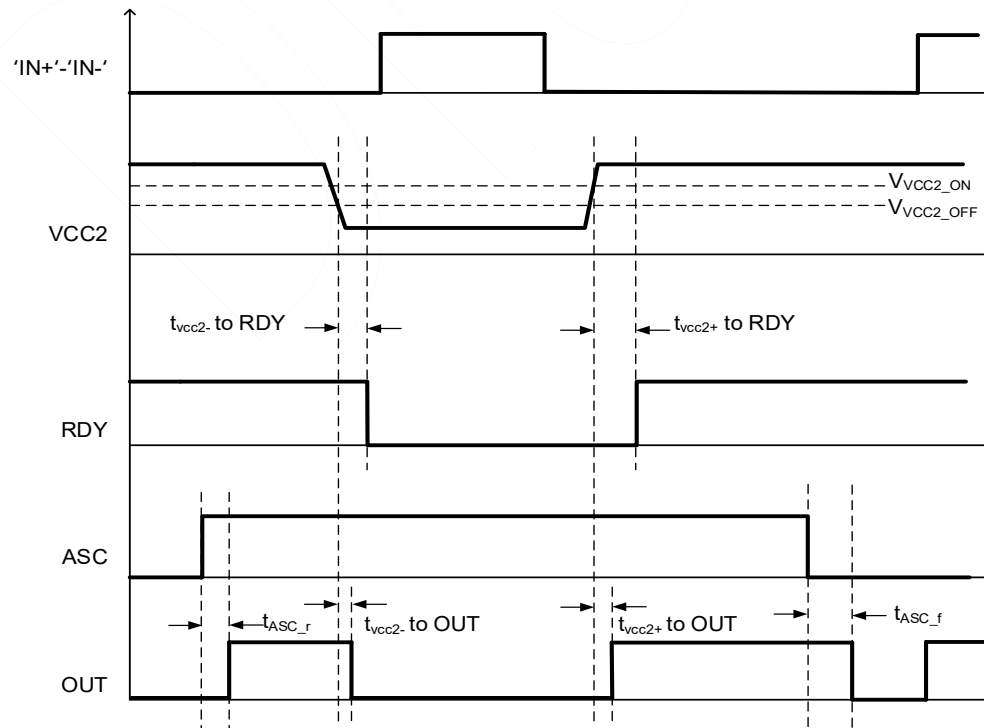


Figure 8.10 ASC protection with V_{CC2} UVLO Timing Diagram

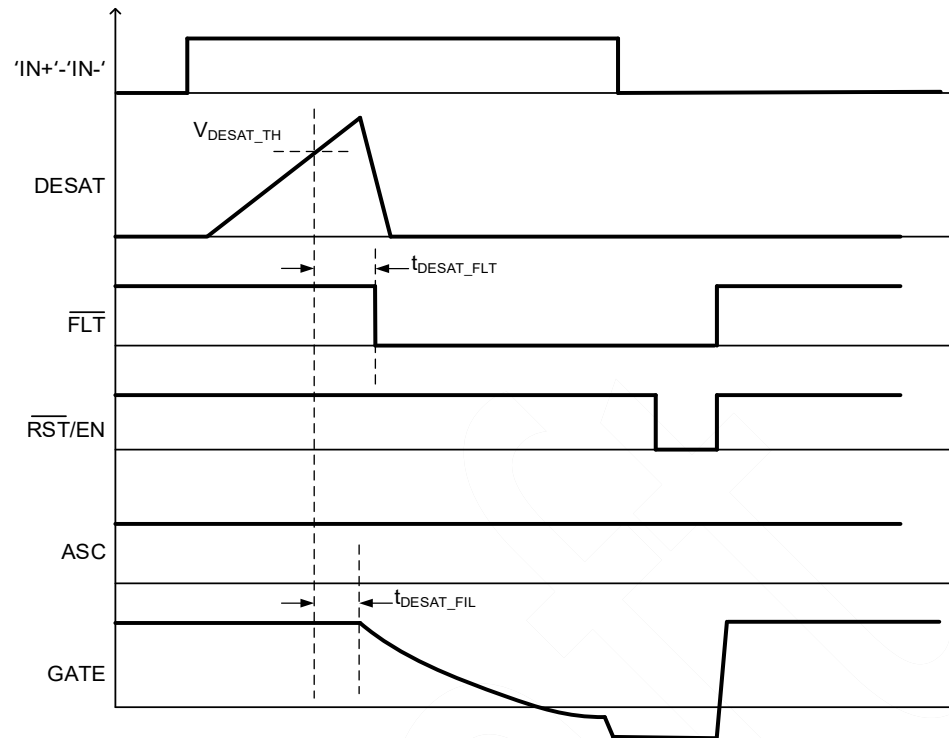


Figure 8.11 ASC protection with DESAT protection Timing Diagram

8.12. Device Functional Modes

Table lists the common functional modes of the device.

Input								Output			
V _{CC1}	V _{CC2}	V _{EE2}	IN+	IN-	EN/RST	DESAT	ASC	RDY	FLT	OUTH/OUTL	CLAMP
PU	PD/OPEN	X	X	X	X	X	X	LOW	HIZ	LOW	LOW
PD	PU	X	X	X	X	X	LOW	LOW	HIZ	LOW	LOW
OPEN	X	X	X	X	X	X	LOW	HIZ	HIZ	LOW	LOW
PU	PU	X	X	X	HIGH	LOW	HIGH	HIZ	HIZ	HIGH	HIZ
PU	PU	X	X	X	LOW	X	HIGH	HIZ	HIZ	HIGH	HIZ
PD/OPEN	PU	X	X	X	X	X	HIGH	X	HIZ	HIGH	HIZ
PU	PU	X	X	X	LOW	X	LOW	HIZ	HIZ	LOW	LOW
PU	PU	X	LOW	X	HIGH	X	LOW	HIZ	HIZ	LOW	LOW
PU	PU	X	HIGH	LOW	HIGH	HIGH	X	HIZ	LOW	LOW	LOW
PU	PU	X	HIGH	LOW	HIGH	LOW	LOW	HIZ	HIZ	HIGH	HIZ
PU	PU	X	HIGH	HIGH	HIGH	X	LOW	HIZ	HIZ	LOW	LOW

Open: V_{CC} < POR ; PU: V_{CC} > V_{CC} UVLO ; PD: POR < V_{CC} < V_{CC} UVLO ; X: Irrelevant ; HIZ :High impedance

POR is around 1.8V.

9.4. Design for Automatic Reset Control

$\overline{\text{RST/EN}}$ pin has two functions. It is used to enable the device and reset the fault signal after DESAT is detected. The $\overline{\text{RST/EN}}$ pin is pulled down to GND by a internal resistor, so the $\overline{\text{RST/EN}}$ pin must be pulled up externally to enable the device.

After DESAT is detected, the FLT pin will be pulled down until the rising edge of the $\overline{\text{RST/EN}}$ pin is coming. To be mentioned, there is a FLT mute time, which means the reset signal must be held for at least $t_{\text{FLT_MUTE}}$.

NSI66x1A can be designed for automatic reset mode. $\overline{\text{RST/EN}}$ pin can be connected with IN+ directly when the PWM is applied to the IN+. Besides, $\overline{\text{RST/EN}}$ pin can be connected with IN- through a NOT logic if the PWM is applied to the IN-. Whichever mode is used, the PWM off time should be longer than the $t_{\text{FLT_MUTE}}$.

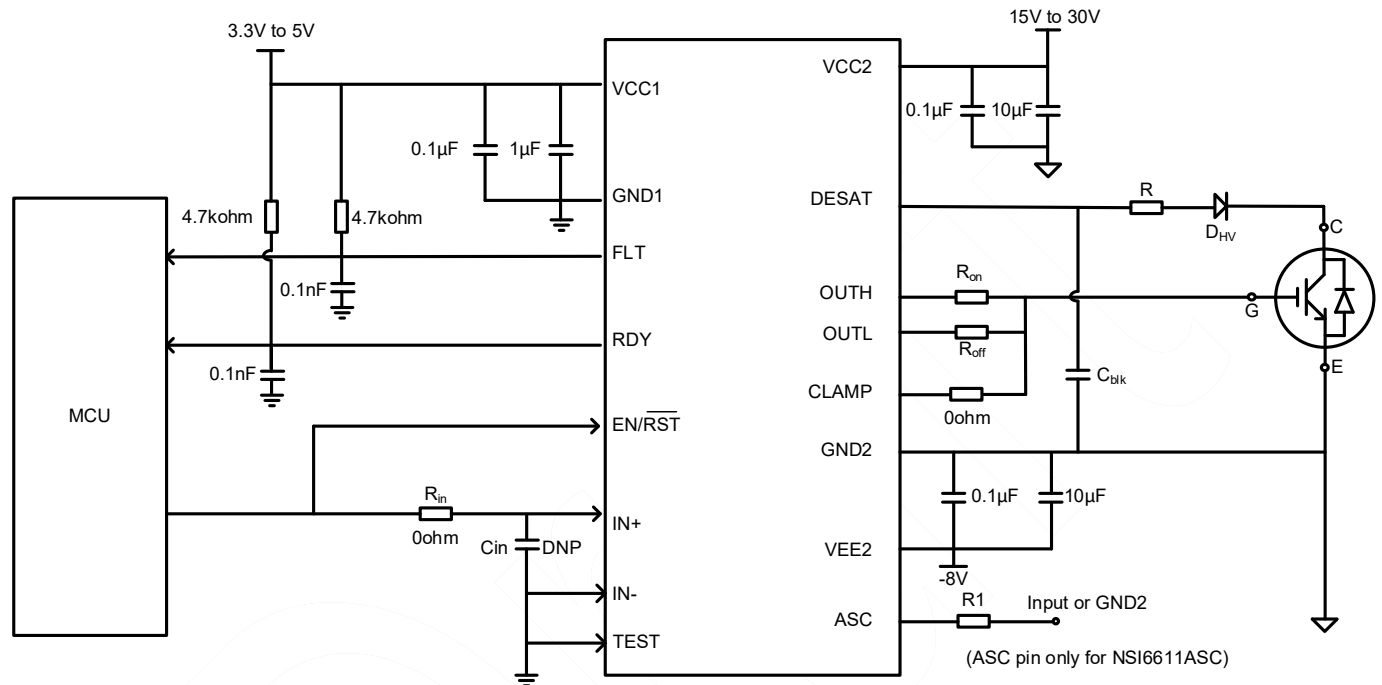


Figure 9.2 Automatic Reset Control (IN+)

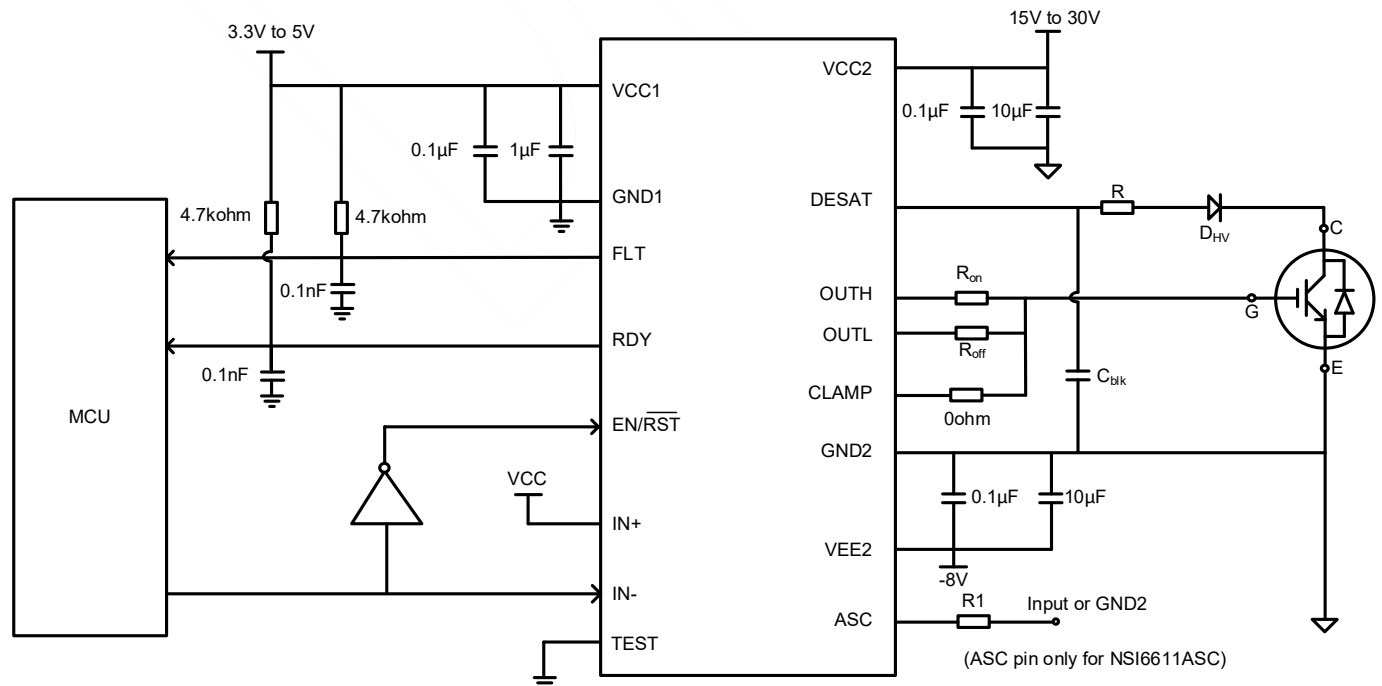


Figure 9.3 Automatic Reset Control (IN-)

9.5. PWM Interlock Protection

For applications to drive power transistors in half bridge configuration, two NSI66x1A can be used. NSI66x1A support Interlock protection. If the controller has some mistakes, leading to negative dead time, the output PWM of NSI66x1A is adjusted to avoid power transistor shoot through.

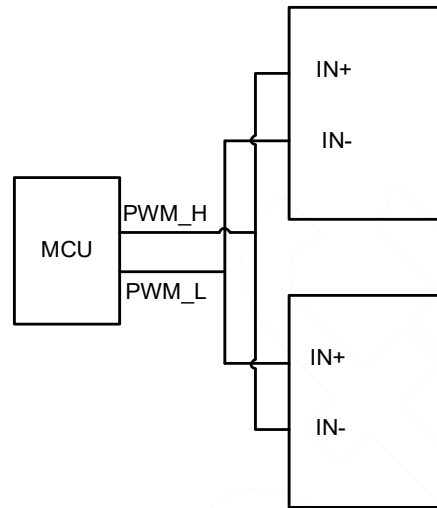


Figure 9.4 Interlock Protection

9.6. Design for R_{on} and R_{off}

NSI66x1A is featured with split output, so the turn on and turn off switching speed can be independently controlled. The turn on and turn off resistance determine the peak source and sink current, which can be estimated by the formula:

$$I_{source} = \min\left(\frac{V_{CC2} - V_{EE}}{R_{ON} + R_{OH} + R_{Gint}}, 10A\right)$$

$$I_{sink} = \min\left(\frac{V_{CC2} - V_{EE}}{R_{OFF} + R_{OL} + R_{Gint}}, 10A\right)$$

Where

R_{Gint} is the internal resistance of the SiC or IGBT.

9.7. Design for DESAT Protection

DESAT is used to protect the power semiconductor from overcurrent. When the voltage of DESAT is over the V_{DESAT_TH} , the block of soft turn off will be activated and the fault pin will be pulled down. For typical application, the crucial components required to build the DESAT circuit are the DESAT diode, DESAT resistor and the blank capacitor.

The DESAT diode function is to conduct forward current. In order to avoid the false detection caused by the reverse recovery spikes, a very fast reverse recovery time diode with small reverse parasitic capacitance is recommended. The DESAT detection threshold voltage of 9V can be reduced by the DESAT diode, which can be calculated as:

$$V_{DESAT} = 9 - V_F$$

The anti-parallel diode of IGBT have a large transient forward voltage of the diode, which may result in a large negative voltage spike on the DESAT pin, then it may draw a large current from driver. DESAT resistor is used to limit the current. A 100Ω resistor is recommended to be added in series with the DESAT diode.

The DESAT fault detection should remain a short blanking time so that the collector voltage can fall below the V_{DESAT_TH} . This blanking time can make sure that there is no nuisance tripping during the IGBT turn-on. It is based on the blank capacitor, which can be estimated as:

$$t_{BLK} = \frac{C_{BLK} \times V_{DESAT_TH}}{I_{CHG}}$$

9.8. Design for External Current Buffer

Totem structure can be used as an external current buffer to increase the IGBT gate drive current, such as the NPN/PNP buffer shown as below. When the external buffer is used, the external components for soft turn off should be designed in addition. The capacitor is used to adjust the timing and the resistor ensure the sink current lower than the I_{OUTL} . Both resistor and capacitor can be estimated by the Equation below.

$$C_{STO} = \frac{I_{STO} \times t_{STO}}{V_{CC2} - V_{EE2}}$$

$$R_{STO} = \frac{V_{CC2} - V_{EE2}}{I_{OUTL}}$$

I_{STO} is the internal soft turn off current

T_{sto} is the expected timing

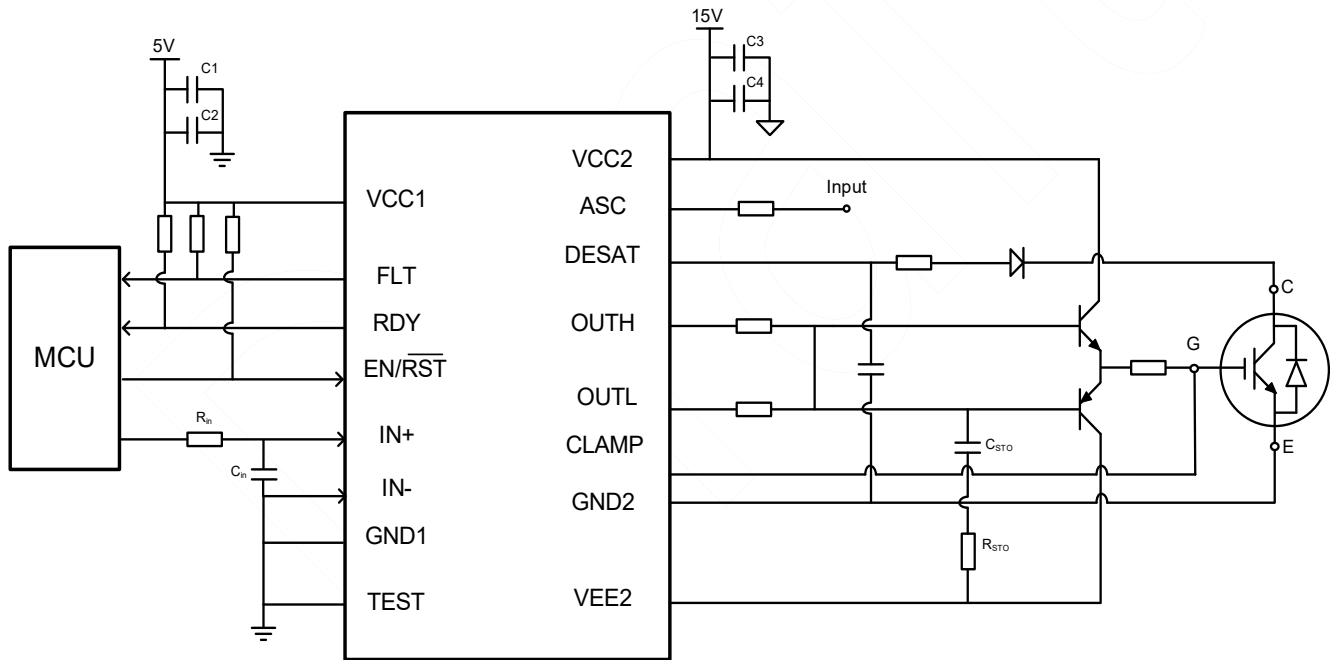


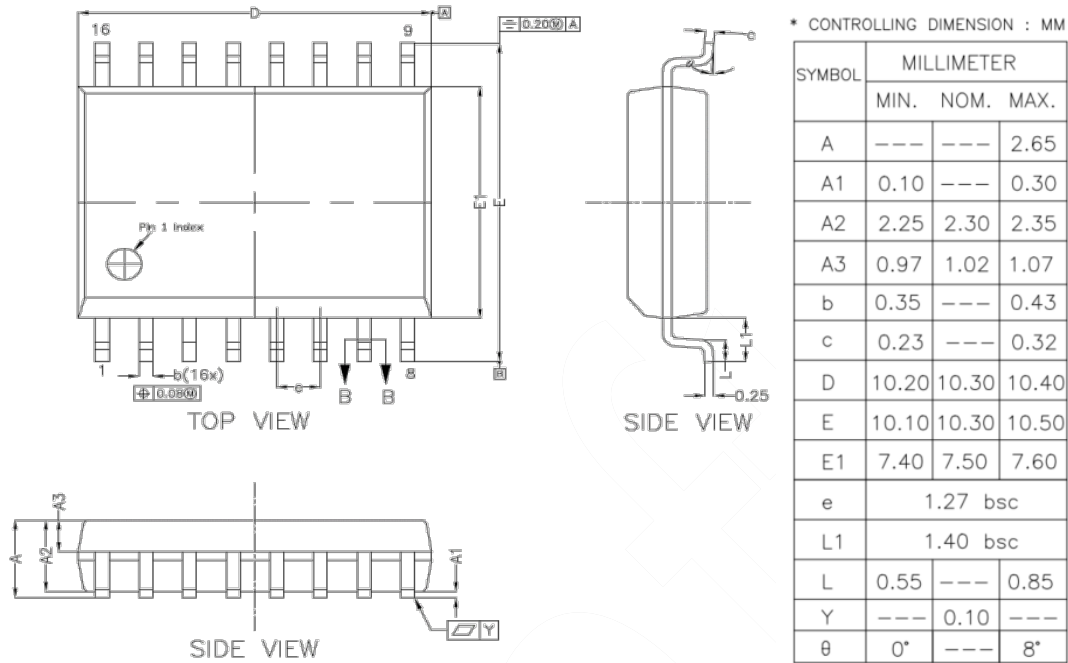
Figure 9.5 External current buffer circuit

9.9. PCB Layout

Careful PCB layout is essential for optimal performance. Some key guidelines are:

- The bypass capacitors should be placed close to NSI66x1A, between V_{CC1} to GND1, V_{CC2} to GND2 and V_{EE2} to GND2.
- There is high switching current that charges and discharges the gate of external power transistor, leading to EMI and ring issues. The parasitic inductance of this loop should be minimized, by decreasing loop area and place NSI66x1A close to power transistor.
- Place large amount of copper connecting to V_{EE2} pin and V_{CC2} pin for thermal dissipation, with priority on V_{EE2} pin. If the system has multi-layers of V_{EE2} or V_{CC2} , use multiple vias of adequate size for connection.
- To ensure isolation performance between primary and secondary side, the space under the chip should keep free from planes, traces, pads or via.

10. Package Information



Dimensions shown in millimeters

Figure 10.1 SOW16 Package Shape and Dimension (ATX)

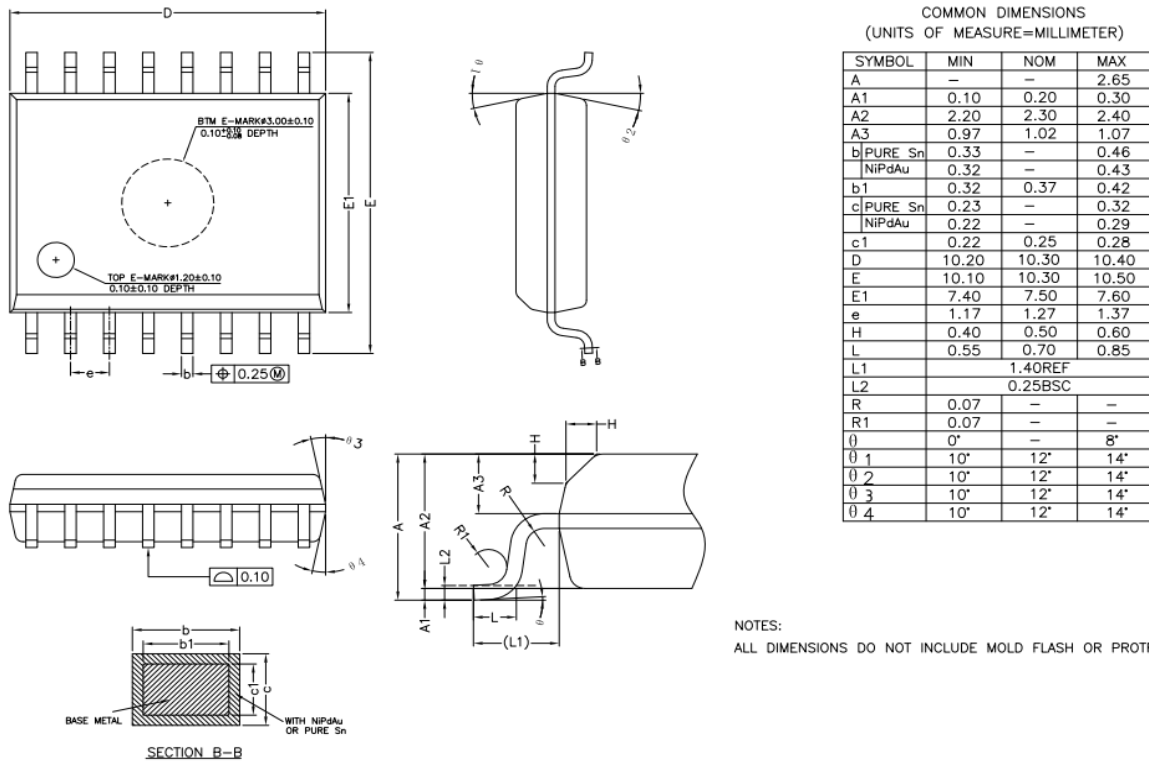


Figure 10.2 SOW16 Package Shape and Dimension (TongFu)

11. Ordering Information

<i>Part Number</i>	<i>ASC Feature</i>	<i>OUT Pin</i>	<i>MSL</i>	<i>Category</i>	<i>Ambient Temperature</i>	<i>MPQ</i>	<i>SPQ</i>	<i>Package Type</i>
NSI6651ASC-Q1SWR	No	OUTH, OUTL	3	Automotive	-40°C to 125°C	1000	1000	SOW16
NSI6651ALC-Q1SWR	No	OUT	3	Automotive	-40°C to 125°C	1000	1000	SOW16
NSI6611ASC-Q1SWR	Yes	OUTH, OUTL	3	Automotive	-40°C to 125°C	1000	1000	SOW16

12. Documentation Support

<i>Part Number</i>	<i>Product Folder</i>	<i>Datasheet</i>	<i>Technical Documents</i>	<i>Isolated Driver Selection Guide</i>
NSI66x1A-Q1	Click here	Click here	Click here	Click here

13. Tape and Reel Information

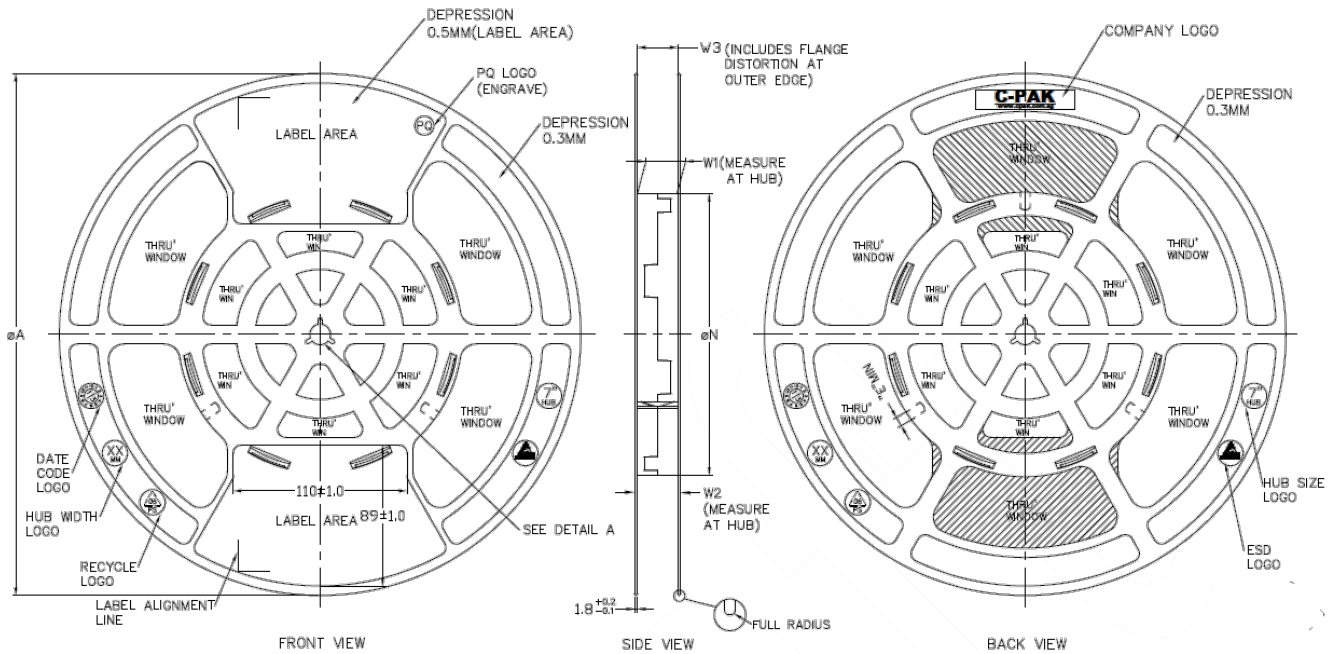
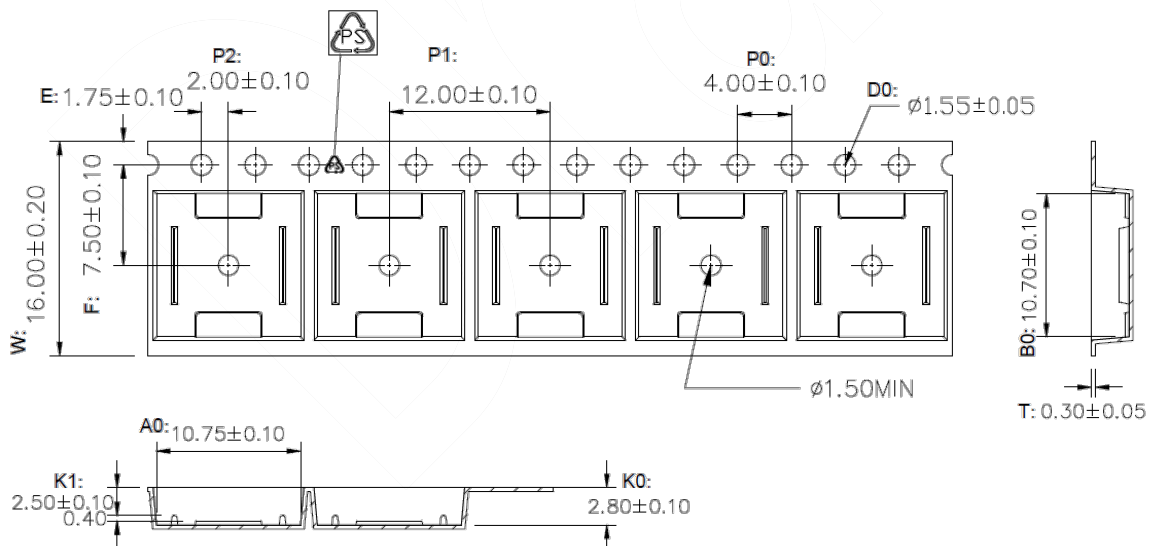


Figure 13.1 Reel Information (ATX)



1. 10 sprocket hole pitch cumulative tolerance ± 0.20 .
2. Carrier camber is within 1 mm in 250 mm.
3. Material : Black Conductive Polystyrene Alloy .
4. All dimensions meet EIA-481 requirements.
5. Thickness : $0.30 \pm 0.05 \text{ mm}$.
6. Packing length per 22" reel : 378 Meters.(N=122)
7. Component load per 13" reel : 1000 pcs.

Figure 13.2 SOW16 Tape Information (ATX)

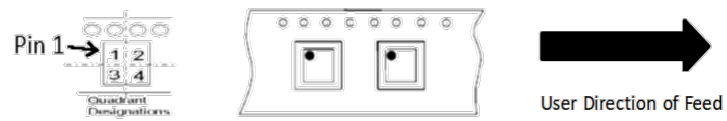


Figure 13.5 Quadrant Designation for Pin1 Orientation in Tape

14. Reversion History

Revision	Description	Date
1.0	Initial version	2023/1/17
1.1	1.Update part number from NSI66x1 to NSI66x1 2. Update device information 3.Update package type from SOP16(300mil) to SOW16 4.Update Safety certification information	2023/9/7
1.2	1. Updated ordering information, which modified the moisture sensitivity level of the SOW16 package from MSL2 to MSL 3 because a new package supplier with MSL3 is added.	2026/3/11

IMPORTANT NOTICE

The information given in this document (the “Document”) shall in no event be regarded as any warranty or authorization of, express or implied, including but not limited to accuracy, completeness, merchantability, fitness for a particular purpose or infringement of any third party’s intellectual property rights.

Users of this Document shall be solely responsible for the use of NOVOSENSE’s products and applications, and for the safety thereof. Users shall comply with all laws, regulations and requirements related to NOVOSENSE’s products and applications, although information or support related to any application may still be provided by NOVOSENSE.

This Document is provided on an “AS IS” basis, and is intended only for skilled developers designing with NOVOSENSE’s products. NOVOSENSE reserves the rights to make corrections, modifications, enhancements, improvements or other changes to the products and services provided without notice. NOVOSENSE authorizes users to use this Document exclusively for the development of relevant applications or systems designed to integrate NOVOSENSE’s products. No license to any intellectual property rights of NOVOSENSE is granted by implication or otherwise. Using this Document for any other purpose, or any unauthorized reproduction or display of this Document is strictly prohibited. In no event shall NOVOSENSE be liable for any claims, damages, costs, losses or liabilities arising out of or in connection with this Document or the use of this Document.

For further information on applications, products and technologies, please contact NOVOSENSE (www.novosns.com).

Suzhou NOVOSENSE Microelectronics Co., Ltd